AWS CLOUD & DATA ENGINEERING

PRACTICAL ASSIGNMENT REPORT

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College	Jain (Deemed-to-be) University
Batch	2023–2027
Region Used	Asia Pacific (Mumbai) — ap-south-1
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AWS S3 • IAM • AWS Glue • Amazon Athena • SQL • Linux

ABSTRACT

This report documents practical work carried out for an AWS Cloud and Data Engineering assignment. The activities evidenced in the supplied materials include SQL practice, Linux command practice, AWS region selection, Amazon S3 bucket and prefix setup, employee CSV data preparation, AWS Glue Data Catalog table creation, Athena queries, and the design of a Glue Studio Visual ETL workflow. The report distinguishes verified successful work from attempts blocked by an AWS account-access denial.

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1. INTRODUCTION AND OBJECTIVES

Cloud data engineering combines cloud storage, cataloguing, transformation, and query services to make datasets available for analysis. This practical assignment uses Amazon S3 as object storage, AWS Glue Data Catalog to describe tabular data, AWS Glue Studio to design a visual ETL pipeline, and Amazon Athena to query data using SQL.

The objectives are to practise SQL and Linux fundamentals; create and organize a small employee dataset in Amazon S3; understand IAM roles and permissions; register a dataset in the Glue Data Catalog; query employee records in Athena; and design a visual ETL workflow that targets curated storage in Parquet format.

2. TOOLS AND ENVIRONMENT

Tool / service	Role in the practical
Amazon S3	Object storage for raw and curated data prefixes.
AWS IAM	Role and policy configuration for Glue access.
AWS Glue Data Catalog	Metadata database and employee table definition.
AWS Glue Studio Visual ETL	Visual data-source, transformation, and target workflow.
Amazon Athena	Serverless SQL queries over catalogued data.
SQL and Linux terminal	Practice of database queries and command-line operations.

3. SQL PRACTICE

The provided evidence includes SQL database/table creation, employee data insertion, filtering, sorting, aggregate calculations, grouping, and salary-related queries. Screenshots are retained in the evidence appendix. Athena results shown in the supplied screenshots include a 10-row employee result set, department employee counts, and a highest-paid employee result.

Representative query patterns supported by the screenshots

Display records
SELECT * FROM employees_new LIMIT 10;

Count employees by department
SELECT department, COUNT(*) AS employee_count FROM employees_new GROUP BY department ORDER BY employee_count DESC;

Find the highest-paid employee
SELECT employee_id, name, department, salary, city FROM employees_new ORDER BY salary DESC LIMIT 1;

The supplied Athena screenshot shows the highest-paid employee as employee_id 107, Arjun, Finance, salary 55000, Bangalore. A department-count result screenshot shows IT 5, Finance 4, Sales 3, and HR 3. These figures are reported as displayed in the available evidence.

4. LINUX PRACTICE

Linux command practice is represented by terminal screenshots covering command groups 01–05, 06–10, and 11–15. These screenshots are included in the appendix as practical evidence. The report does not claim commands beyond those evidenced by the supplied images.

5. AWS ACCOUNT AND REGION

The AWS console screenshots show the Asia Pacific (Mumbai) region, ap-south-1. This region selection matters because AWS resources and data locations may be region-specific. The report records the visible region and does not include passwords, access keys, secret keys, or other credentials.

6. AMAZON S3 DATA LAKE SETUP

The S3 bucket shown in the supplied evidence is deekshith-aws-data-engineering-assignment. The bucket contains raw/ and curated/ prefixes. The employee CSV file employees (1).csv is shown in the raw/ and curated/ locations. The raw prefix represents the uploaded source data; curated/ is intended to hold transformed output. A CSV file manually uploaded to curated/ is not, by itself, proof that the ETL job generated Parquet output.

7. IAM ROLE AND PERMISSIONS

The Glue role visible in the screenshots is AWSGlueServiceRole-EmployeeData. IAM roles provide AWS services with temporary permissions to access resources. The role and its attached permissions are documented as observed; no secrets or credentials are reproduced in this report.

8. AWS GLUE DATA CATALOG AND TABLE

The database shown in the practical evidence is employee_data_db. A table named employees_new was successfully created manually in the Data Catalog, with its location pointing to the raw/ S3 prefix and CSV classification. The visible schema fields include employee_id, name, department, salary, and city. The task checklist names glue_training_db, while the screenshots show employee_data_db; this report follows the screenshots for the implemented work and notes the discrepancy rather than silently reconciling it.

9. AMAZON ATHENA QUERYING

Athena was used to query the catalogued employee data. The supplied results include employee rows, a grouped department-count result, and the top salary result. An initial parsing error was also observed when a header value employee_id was parsed as an integer. This commonly occurs when a CSV header is treated as a data row. The later screenshots show successful query results, so the report records both the error and the successful results without asserting an undocumented fix.

10. AWS GLUE STUDIO VISUAL ETL

The visual design shown in the screenshots contains an AWS Glue Data Catalog source, a Drop Fields transformation, and an Amazon S3 target. The target settings show Parquet format and Snappy compression. The intended workflow is Data Catalog → Drop Fields → S3 target. The visible transformation panel lists employee_id, name, department, salary, and city; the screenshots do not conclusively establish that the job was successfully saved or executed.

11. ERRORS, TROUBLESHOOTING, AND LIMITATIONS

AWS Glue crawler creation and Glue Studio job creation/update attempts displayed an AccessDeniedException stating that account 171665661965 is denied access. This means the requested operation was rejected; the screenshots do not establish whether the cause is an account-level restriction, an IAM permission boundary, a service restriction, or another authorization control. The same denial was reported for a friend's account, but that alone does not prove a Free Tier limitation.

The Glue crawler and ETL job must therefore be marked **attempted but unsuccessful** based on the available evidence. Do not claim a successful crawler run, a succeeded ETL job, or generated Parquet files unless separate evidence confirms these outcomes. To resolve the denial, an AWS account administrator or training-lab provider may need to inspect CloudTrail event details and account-level permissions.

12. RESULTS AND CONCLUSION

The evidence confirms that the employee CSV was uploaded to S3, a Glue Data Catalog database and table were created, and Athena successfully returned employee records and aggregate query results. SQL and Linux practice screenshots are also available. A Glue Studio ETL graph was configured visually, but the attempt to save/create the job was blocked by an account-access denial. The assignment report therefore separates completed work from the blocked Glue execution stages.

The practical demonstrates the core data-engineering flow: S3 stores the data, Glue Data Catalog records its structure, Athena queries the table using SQL, and Glue Studio provides a visual method to design transformations and write curated data. Successful end-to-end ETL completion remains unverified due to the access-denied error.

13. REFERENCES

1. AWS Task Checklist supplied with the assignment.
2. Amazon S3 documentation, https://docs.aws.amazon.com/AmazonS3/latest/userguide/Welcome.html
3. AWS Glue documentation, https://docs.aws.amazon.com/glue/
4. Amazon Athena documentation, https://docs.aws.amazon.com/athena/
5. AWS IAM documentation, https://docs.aws.amazon.com/iam/

APPENDIX A. SCREENSHOT EVIDENCE

The following appendix includes 58 screenshot images available in the current workspace, each with a numbered caption. The original screenshot inventory visible in the user's Finder also included filenames not all separately mounted here; see the delivery note for that limitation.

SCREENSHOT EVIDENCE 01 OF 58

```
mkdir -p ~/AWS_Cloud_Data_Engineering/{SQL,Linux,Screenshots,Report,Data}
cd ~/AWS_Cloud_Data_Engineering
pwd
ls

/Users/nandithagn/AWS_Cloud_Data_Engineering
Data            Linux           Report          Screenshots     SQL
nandithagn@Nandithas-MacBook-Air AWS_Cloud_Data_Engineering % X#
nandithagn@Nandithas-MacBook-Air AWS_Cloud_Data_Engineering %
nandithagn@Nandithas-MacBook-Air AWS_Cloud_Data_Engineering %
mysql --version

mysql  Ver 26.7.0 for macos26.6 on arm64 (Homebrew)
nandithagn@Nandithas-MacBook-Air AWS_Cloud_Data_Engineering %
brew services list

Name  Status User       File
mysql other  nandithagn ~/Library/LaunchAgents/sh.brew.mysql.plist
nandithagn@Nandithas-MacBook-Air AWS_Cloud_Data_Engineering %
mysqladmin ping

mysqladmin: connect to server at 'localhost' failed
error: 'Access denied for user 'nandithagn'@'localhost' (using password: NO)'
nandithagn@Nandithas-MacBook-Air AWS_Cloud_Data_Engineering %
mysql -u root -p

Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 19
Server version: 8.4.11 MySQL Community Server - GPL

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Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
mysql> CREATE DATABASE employee_training_db;
Query OK, 1 row affected (0.037 sec)

mysql>
mysql> USE employee_training_db;
Database changed
mysql>
mysql>
mysql> SHOW DATABASES;
+--------------------------+
| Database                 |
+--------------------------+
| college                  |
| ecommerce                |
| employee_training_db     |
| information_schema       |
| junior_data_engineering  |
| mysql                    |
| performance_schema       |
| sys                      |
+--------------------------+
8 rows in set (0.001 sec)

mysql>
```

Figure 1: 06 database creation

Source file: 06_database_creation.png

SCREENSHOT EVIDENCE 02 OF 58

```
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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
mysql> CREATE DATABASE employee_training_db;
Query OK, 1 row affected (0.037 sec)

mysql>
mysql> USE employee_training_db;
Database changed
mysql>
mysql>
mysql> SHOW DATABASES;
+----------------------------+
| Database                   |
+----------------------------+
| college                    |
| ecommerce                  |
| employee_training_db       |
| information_schema         |
| junior_data_engineering    |
| mysql                      |
| performance_schema         |
| sys                        |
+----------------------------+
8 rows in set (0.001 sec)

mysql>
mysql> CREATE TABLE employees (
    ->     employee_id INT PRIMARY KEY,
    ->     name VARCHAR(50),
    ->     department VARCHAR(30),
    ->     salary DECIMAL(10,2),
    ->     city VARCHAR(30)
    -> );
Query OK, 0 rows affected (0.036 sec)

mysql>
mysql> SHOW TABLES;
+----------------------------------+
| Tables_in_employee_training_db |
+----------------------------------+
| employees                        |
+----------------------------------+
1 row in set (0.002 sec)

mysql>
mysql>
mysql> DESCRIBE employees;
+-------------+--------------+------+-----+---------+-------+
| Field       | Type         | Null | Key | Default | Extra |
+-------------+--------------+------+-----+---------+-------+
| employee_id | int          | NO   | PRI | NULL    |       |
| name        | varchar(50)  | YES  |     | NULL    |       |
| department  | varchar(30)  | YES  |     | NULL    |       |
| salary      | decimal(10,2)| YES  |     | NULL    |       |
| city        | varchar(30)  | YES  |     | NULL    |       |
+-------------+--------------+------+-----+---------+-------+
5 rows in set (0.006 sec)

mysql>
```

Figure 2: 07 employees table creation

Source file: 07_employees_table_creation.png

SCREENSHOT EVIDENCE 03 OF 58

```
| Field       | Type          | Null | Key | Default | Extra |
+-------------+---------------+------+-----+---------+-------+
| employee_id | int           | NO   | PRI | NULL    |       |
| name        | varchar(50)   | YES  |     | NULL    |       |
| department  | varchar(30)   | YES  |     | NULL    |       |
| salary      | decimal(10,2) | YES  |     | NULL    |       |
| city        | varchar(30)   | YES  |     | NULL    |       |
+-------------+---------------+------+-----+---------+-------+
5 rows in set (0.006 sec)

mysql>
mysql> INSERT INTO employees (employee_id, name, department, salary, city)
    -> VALUES
    -> (101, 'Aarav', 'IT', 45000.00, 'Bangalore'),
    -> (102, 'Diya', 'HR', 38000.00, 'Mysore'),
    -> (103, 'Rohan', 'Finance', 52000.00, 'Hubli'),
    -> (104, 'Ananya', 'IT', 48000.00, 'Bangalore'),
    -> (105, 'Vikram', 'Sales', 35000.00, 'Mangalore'),
    -> (106, 'Meera', 'HR', 42000.00, 'Hassan'),
    -> (107, 'Arjun', 'Finance', 55000.00, 'Bangalore'),
    -> (108, 'Kavya', 'IT', 50000.00, 'Mysore'),
    -> (109, 'Rahul', 'Sales', 39000.00, 'Hubli'),
    -> (110, 'Sneha', 'Finance', 47000.00, 'Hassan'),
    -> (111, 'Kiran', 'IT', 46000.00, 'Mangalore'),
    -> (112, 'Neha', 'HR', 40000.00, 'Bangalore'),
    -> (113, 'Manoj', 'Sales', 36000.00, 'Mysore'),
    -> (114, 'Pooja', 'Finance', 53000.00, 'Hubli'),
    -> (115, 'Aditya', 'IT', 49000.00, 'Hassan');
Query OK, 15 rows affected (0.006 sec)
Records: 15  Duplicates: 0  Warnings: 0

mysql>
mysql> SELECT COUNT(*) AS total_employees FROM employees;
+-----------------+
| total_employees |
+-----------------+
|              15 |
+-----------------+
1 row in set (0.008 sec)

mysql>
mysql> SELECT * FROM employees;
+-------------+--------+------------+----------+-----------+
| employee_id | name   | department | salary   | city      |
+-------------+--------+------------+----------+-----------+
|         101 | Aarav  | IT         | 45000.00 | Bangalore |
|         102 | Diya   | HR         | 38000.00 | Mysore    |
|         103 | Rohan  | Finance    | 52000.00 | Hubli     |
|         104 | Ananya | IT         | 48000.00 | Bangalore |
|         105 | Vikram | Sales      | 35000.00 | Mangalore |
|         106 | Meera  | HR         | 42000.00 | Hassan    |
|         107 | Arjun  | Finance    | 55000.00 | Bangalore |
|         108 | Kavya  | IT         | 50000.00 | Mysore    |
|         109 | Rahul  | Sales      | 39000.00 | Hubli     |
|         110 | Sneha  | Finance    | 47000.00 | Hassan    |
|         111 | Kiran  | IT         | 46000.00 | Mangalore |
|         112 | Neha   | HR         | 40000.00 | Bangalore |
|         113 | Manoj  | Sales      | 36000.00 | Mysore    |
|         114 | Pooja  | Finance    | 53000.00 | Hubli     |
|         115 | Aditya | IT         | 49000.00 | Hassan    |
+-------------+--------+------------+----------+-----------+
15 rows in set (0.002 sec)

mysql>
```

Figure 3: 08 employee data inserted

Source file: 08_employee_data_inserted.png

SCREENSHOT EVIDENCE 04 OF 58

```
    -> (102, 'Diya', 'HR', 38000.00, 'Mysore'),
    -> (103, 'Rohan', 'Finance', 52000.00, 'Hubli'),
    -> (104, 'Ananya', 'IT', 48000.00, 'Bangalore'),
    -> (105, 'Vikram', 'Sales', 35000.00, 'Mangalore'),
    -> (106, 'Meera', 'HR', 42000.00, 'Hassan'),
    -> (107, 'Arjun', 'Finance', 55000.00, 'Bangalore'),
    -> (108, 'Kavya', 'IT', 50000.00, 'Mysore'),
    -> (109, 'Rahul', 'Sales', 39000.00, 'Hubli'),
    -> (110, 'Sneha', 'Finance', 47000.00, 'Hassan'),
    -> (111, 'Kiran', 'IT', 46000.00, 'Mangalore'),
    -> (112, 'Neha', 'HR', 40000.00, 'Bangalore'),
    -> (113, 'Manoj', 'Sales', 36000.00, 'Mysore'),
    -> (114, 'Pooja', 'Finance', 53000.00, 'Hubli'),
    -> (115, 'Aditya', 'IT', 49000.00, 'Hassan');
Query OK, 15 rows affected (0.006 sec)
Records: 15  Duplicates: 0  Warnings: 0

mysql>
mysql> SELECT COUNT(*) AS total_employees FROM employees;
+-----------------+
| total_employees |
+-----------------+
|              15 |
+-----------------+
1 row in set (0.008 sec)

mysql>
mysql> SELECT * FROM employees;
+-------------+--------+------------+----------+-----------+
| employee_id | name   | department | salary   | city      |
+-------------+--------+------------+----------+-----------+
|         101 | Aarav  | IT         | 45000.00 | Bangalore |
|         102 | Diya   | HR         | 38000.00 | Mysore    |
|         103 | Rohan  | Finance    | 52000.00 | Hubli     |
|         104 | Ananya | IT         | 48000.00 | Bangalore |
|         105 | Vikram | Sales      | 35000.00 | Mangalore |
|         106 | Meera  | HR         | 42000.00 | Hassan    |
|         107 | Arjun  | Finance    | 55000.00 | Bangalore |
|         108 | Kavya  | IT         | 50000.00 | Mysore    |
|         109 | Rahul  | Sales      | 39000.00 | Hubli     |
|         110 | Sneha  | Finance    | 47000.00 | Hassan    |
|         111 | Kiran  | IT         | 46000.00 | Mangalore |
|         112 | Neha   | HR         | 40000.00 | Bangalore |
|         113 | Manoj  | Sales      | 36000.00 | Mysore    |
|         114 | Pooja  | Finance    | 53000.00 | Hubli     |
|         115 | Aditya | IT         | 49000.00 | Hassan    |
+-------------+--------+------------+----------+-----------+
15 rows in set (0.002 sec)

mysql>
mysql> SELECT * FROM employees
    -> WHERE department = 'IT';
+-------------+--------+------------+----------+-----------+
| employee_id | name   | department | salary   | city      |
+-------------+--------+------------+----------+-----------+
|         101 | Aarav  | IT         | 45000.00 | Bangalore |
|         104 | Ananya | IT         | 48000.00 | Bangalore |
|         108 | Kavya  | IT         | 50000.00 | Mysore    |
|         111 | Kiran  | IT         | 46000.00 | Mangalore |
|         115 | Aditya | IT         | 49000.00 | Hassan    |
+-------------+--------+------------+----------+-----------+
5 rows in set (0.003 sec)

mysql>
```

Figure 4: 09 sql query 03 where it

Source file: 09_sql_query_03_where_it.png

SCREENSHOT EVIDENCE 05 OF 58

```
mysql>
mysql> SELECT COUNT(*) AS total_employees FROM employees;
+-----------------+
| total_employees |
+-----------------+
|              15 |
+-----------------+
1 row in set (0.008 sec)

mysql>
mysql> SELECT * FROM employees;
+-------------+--------+------------+----------+-----------+
| employee_id | name   | department | salary   | city      |
+-------------+--------+------------+----------+-----------+
|         101 | Aarav  | IT         | 45000.00 | Bangalore |
|         102 | Diya   | HR         | 38000.00 | Mysore    |
|         103 | Rohan  | Finance    | 52000.00 | Hubli     |
|         104 | Ananya | IT         | 48000.00 | Bangalore |
|         105 | Vikram | Sales      | 35000.00 | Mangalore |
|         106 | Meera  | HR         | 42000.00 | Hassan    |
|         107 | Arjun  | Finance    | 55000.00 | Bangalore |
|         108 | Kavya  | IT         | 50000.00 | Mysoro    |
|         109 | Rahul  | Sales      | 39000.00 | Hubli     |
|         110 | Sneha  | Finance    | 47000.00 | Hassan    |
|         111 | Kiran  | IT         | 46000.00 | Mangalore |
|         112 | Neha   | HR         | 40000.00 | Bangalore |
|         113 | Manoj  | Sales      | 36000.00 | Mysore    |
|         114 | Pooja  | Finance    | 53000.00 | Hubli     |
|         115 | Aditya | IT         | 49000.00 | Hassan    |
+-------------+--------+------------+----------+-----------+
15 rows in set (0.002 sec)

mysql>
mysql> SELECT * FROM employees
    -> WHERE department = 'IT';
+-------------+--------+------------+----------+-----------+
| employee_id | name   | department | salary   | city      |
+-------------+--------+------------+----------+-----------+
|         101 | Aarav  | IT         | 45000.00 | Bangalore |
|         104 | Ananya | IT         | 48000.00 | Bangalore |
|         108 | Kavya  | IT         | 50000.00 | Mysore    |
|         111 | Kiran  | IT         | 46000.00 | Mangalore |
|         115 | Aditya | IT         | 49000.00 | Hassan    |
+-------------+--------+------------+----------+-----------+
5 rows in set (0.003 sec)

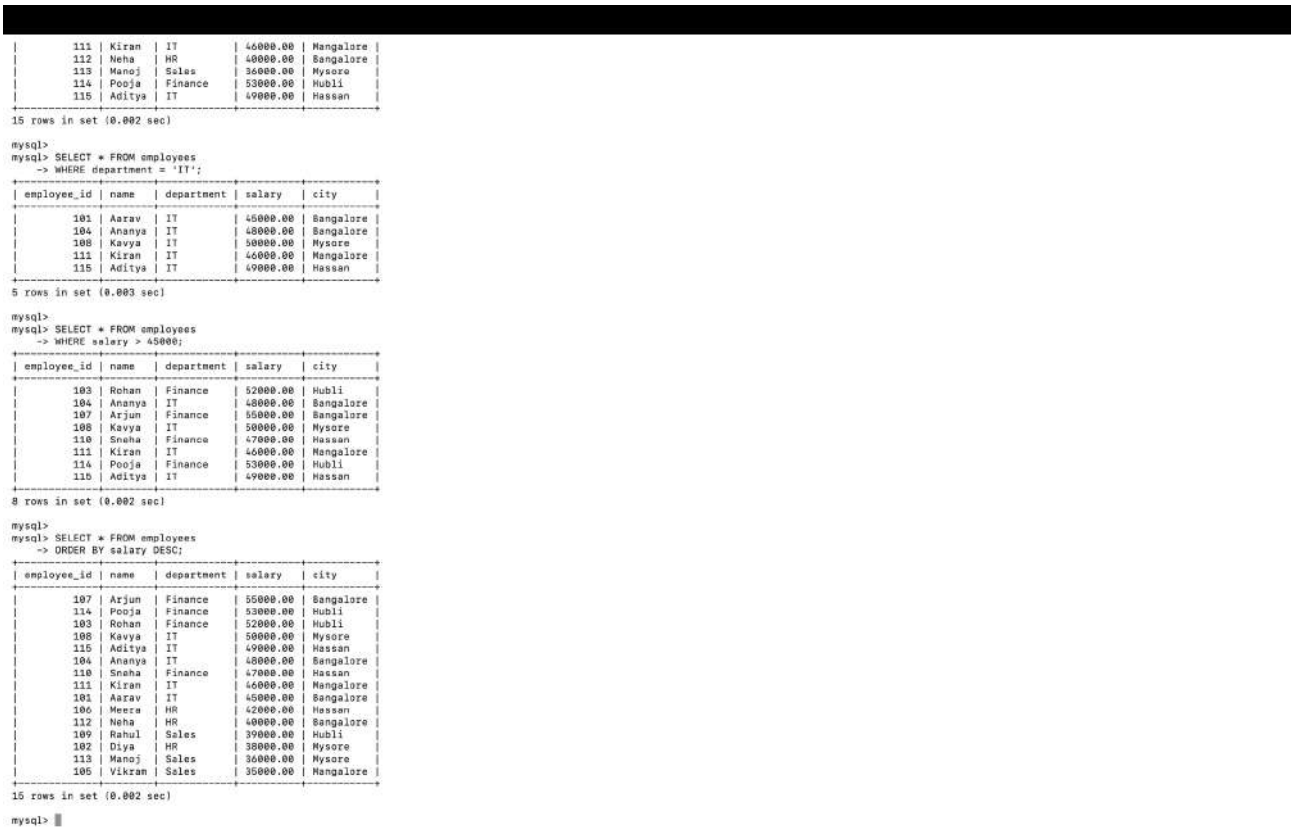
mysql>
mysql> SELECT * FROM employees
    -> WHERE salary > 45000;
+-------------+--------+------------+----------+-----------+
| employee_id | name   | department | salary   | city      |
+-------------+--------+------------+----------+-----------+
|         103 | Rohan  | Finance    | 52000.00 | Hubli     |
|         104 | Ananya | IT         | 48000.00 | Bangalore |
|         107 | Arjun  | Finance    | 55000.00 | Bangalore |
|         108 | Kavya  | IT         | 50000.00 | Mysore    |
|         110 | Sneha  | Finance    | 47000.00 | Hassan    |
|         111 | Kiran  | IT         | 46000.00 | Mangalore |
|         114 | Pooja  | Finance    | 53000.00 | Hubli     |
|         115 | Aditya | IT         | 49000.00 | Hassan    |
+-------------+--------+------------+----------+-----------+
8 rows in set (0.002 sec)

mysql>
```

Figure 5: 10 sql query 04 salary filter

Source file: 10_sql_query_04_salary_filter.png

SCREENSHOT EVIDENCE 06 OF 58



```
|         111 | Kiran   | IT          | 46000.00 | Mangalore |
|         112 | Neha    | HR          | 40000.00 | Bangalore |
|         113 | Manoj   | Sales       | 36000.00 | Mysore    |
|         114 | Pooja   | Finance     | 53000.00 | Hubli     |
|         115 | Aditya  | IT          | 49000.00 | Hassan    |
+-------------+---------+-------------+----------+-----------+
15 rows in set (0.002 sec)

mysql>
mysql> SELECT * FROM employees
    -> WHERE department = 'IT';
+-------------+---------+-------------+----------+-----------+
| employee_id | name    | department  | salary   | city      |
+-------------+---------+-------------+----------+-----------+
|         101 | Aarav   | IT          | 45000.00 | Bangalore |
|         104 | Ananya  | IT          | 48000.00 | Bangalore |
|         108 | Kavya   | IT          | 50000.00 | Mysore    |
|         111 | Kiran   | IT          | 46000.00 | Mangalore |
|         115 | Aditya  | IT          | 49000.00 | Hassan    |
+-------------+---------+-------------+----------+-----------+
5 rows in set (0.003 sec)

mysql>
mysql> SELECT * FROM employees
    -> WHERE salary > 45000;
+-------------+---------+-------------+----------+-----------+
| employee_id | name    | department  | salary   | city      |
+-------------+---------+-------------+----------+-----------+
|         103 | Rohan   | Finance     | 52000.00 | Hubli     |
|         104 | Ananya  | IT          | 48000.00 | Bangalore |
|         107 | Arjun   | Finance     | 55000.00 | Bangalore |
|         108 | Kavya   | IT          | 50000.00 | Mysore    |
|         110 | Sneha   | Finance     | 47000.00 | Hassan    |
|         111 | Kiran   | IT          | 46000.00 | Mangalore |
|         114 | Pooja   | Finance     | 53000.00 | Hubli     |
|         115 | Aditya  | IT          | 49000.00 | Hassan    |
+-------------+---------+-------------+----------+-----------+
8 rows in set (0.002 sec)

mysql>
mysql> SELECT * FROM employees
    -> ORDER BY salary DESC;
+-------------+---------+-------------+----------+-----------+
| employee_id | name    | department  | salary   | city      |
+-------------+---------+-------------+----------+-----------+
|         107 | Arjun   | Finance     | 55000.00 | Bangalore |
|         114 | Pooja   | Finance     | 53000.00 | Hubli     |
|         103 | Rohan   | Finance     | 52000.00 | Hubli     |
|         108 | Kavya   | IT          | 50000.00 | Mysore    |
|         115 | Aditya  | IT          | 49000.00 | Hassan    |
|         104 | Ananya  | IT          | 48000.00 | Bangalore |
|         110 | Sneha   | Finance     | 47000.00 | Hassan    |
|         111 | Kiran   | IT          | 46000.00 | Mangalore |
|         101 | Aarav   | IT          | 45000.00 | Bangalore |
|         106 | Meera   | HR          | 42000.00 | Hassan    |
|         112 | Neha    | HR          | 40000.00 | Bangalore |
|         109 | Rahul   | Sales       | 39000.00 | Hubli     |
|         102 | Diya    | HR          | 38000.00 | Mysore    |
|         113 | Manoj   | Sales       | 36000.00 | Mysore    |
|         105 | Vikram  | Sales       | 35000.00 | Mangalore |
+-------------+---------+-------------+----------+-----------+
15 rows in set (0.002 sec)

mysql>
```

Figure 6: 11 sql query 05 order by

Source file: 11_sql_query_05_order_by.png

SCREENSHOT EVIDENCE 07 OF 58

```
5 rows in set (0.003 sec)

mysql>
mysql> SELECT * FROM employees
    -> WHERE salary > 45000;
+-------------+--------+------------+----------+-----------+
| employee_id | name   | department | salary   | city      |
+-------------+--------+------------+----------+-----------+
|         103 | Rohan  | Finance    | 52000.00 | Hubli     |
|         104 | Ananya | IT         | 48000.00 | Bangalore |
|         107 | Arjun  | Finance    | 55000.00 | Bangalore |
|         108 | Kavya  | IT         | 50000.00 | Mysore    |
|         110 | Sneha  | Finance    | 47000.00 | Hassan    |
|         111 | Kiran  | IT         | 46000.00 | Mangalore |
|         114 | Pooja  | Finance    | 53000.00 | Hubli     |
|         115 | Aditya | IT         | 49000.00 | Hassan    |
+-------------+--------+------------+----------+-----------+
8 rows in set (0.002 sec)

mysql>
mysql> SELECT * FROM employees
    -> ORDER BY salary DESC;
+-------------+--------+------------+----------+-----------+
| employee_id | name   | department | salary   | city      |
+-------------+--------+------------+----------+-----------+
|         107 | Arjun  | Finance    | 55000.00 | Bangalore |
|         114 | Pooja  | Finance    | 53000.00 | Hubli     |
|         103 | Rohan  | Finance    | 52000.00 | Hubli     |
|         108 | Kavya  | IT         | 50000.00 | Mysore    |
|         115 | Aditya | IT         | 49000.00 | Hassan    |
|         104 | Ananya | IT         | 48000.00 | Bangalore |
|         110 | Sneha  | Finance    | 47000.00 | Hassan    |
|         111 | Kiran  | IT         | 46000.00 | Mangalore |
|         101 | Aarav  | IT         | 45000.00 | Bangalore |
|         106 | Meera  | HR         | 42000.00 | Hassan    |
|         112 | Neha   | HR         | 40000.00 | Bangalore |
|         109 | Rahul  | Sales      | 39000.00 | Hubli     |
|         102 | Diya   | HR         | 38000.00 | Mysore    |
|         113 | Manoj  | Sales      | 36000.00 | Mysore    |
|         105 | Vikram | Sales      | 35000.00 | Mangalore |
+-------------+--------+------------+----------+-----------+
15 rows in set (0.002 sec)

mysql>
mysql> SELECT MAX(salary) AS highest_salary
    -> FROM employees;
+----------------+
| highest_salary |
+----------------+
|       55000.00 |
+----------------+
1 row in set (0.003 sec)

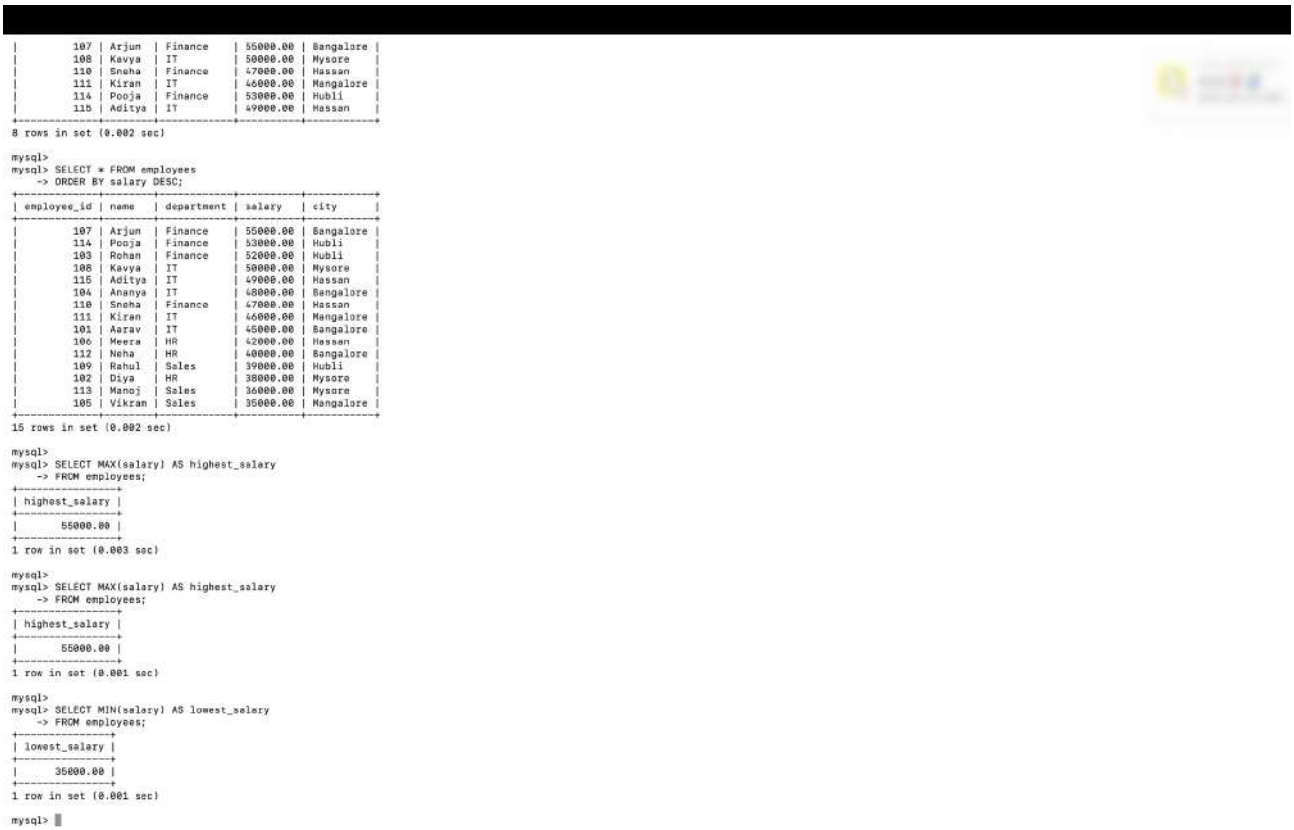
mysql>
mysql> SELECT MAX(salary) AS highest_salary
    -> FROM employees;
+----------------+
| highest_salary |
+----------------+
|       55000.00 |
+----------------+
1 row in set (0.001 sec)

mysql>
```

Figure 7: 12 sql query 06 max salary

Source file: 12_sql_query_06_max_salary.png

SCREENSHOT EVIDENCE 08 OF 58



```
|         107 | Arjun   | Finance    | 55000.00 | Bangalore |
|         108 | Kavya   | IT         | 50000.00 | Mysore    |
|         110 | Sneha   | Finance    | 47000.00 | Hassan    |
|         111 | Kiran   | IT         | 46000.00 | Mangalore |
|         114 | Pooja   | Finance    | 53000.00 | Hubli     |
|         115 | Aditya  | IT         | 49000.00 | Hassan    |
+-------------+---------+------------+----------+-----------+
8 rows in set (0.002 sec)

mysql>
mysql> SELECT * FROM employees
    -> ORDER BY salary DESC;
+-------------+---------+------------+----------+-----------+
| employee_id | name    | department | salary   | city      |
+-------------+---------+------------+----------+-----------+
|         107 | Arjun   | Finance    | 55000.00 | Bangalore |
|         114 | Pooja   | Finance    | 53000.00 | Hubli     |
|         103 | Rohan   | Finance    | 52000.00 | Hubli     |
|         108 | Kavya   | IT         | 50000.00 | Mysore    |
|         115 | Aditya  | IT         | 49000.00 | Hassan    |
|         104 | Ananya  | IT         | 48000.00 | Bangalore |
|         110 | Sneha   | Finance    | 47000.00 | Hassan    |
|         111 | Kiran   | IT         | 46000.00 | Mangalore |
|         101 | Aarav   | IT         | 45000.00 | Bangalore |
|         106 | Meera   | HR         | 42000.00 | Hassan    |
|         112 | Neha    | HR         | 40000.00 | Bangalore |
|         109 | Rahul   | Sales      | 39000.00 | Hubli     |
|         102 | Diya    | HR         | 38000.00 | Mysore    |
|         113 | Manoj   | Sales      | 36000.00 | Mysore    |
|         105 | Vikram  | Sales      | 35000.00 | Mangalore |
+-------------+---------+------------+----------+-----------+
15 rows in set (0.002 sec)

mysql>
mysql> SELECT MAX(salary) AS highest_salary
    -> FROM employees;
+----------------+
| highest_salary |
+----------------+
|       55000.00 |
+----------------+
1 row in set (0.003 sec)

mysql>
mysql> SELECT MAX(salary) AS highest_salary
    -> FROM employees;
+----------------+
| highest_salary |
+----------------+
|       55000.00 |
+----------------+
1 row in set (0.001 sec)

mysql>
mysql> SELECT MIN(salary) AS lowest_salary
    -> FROM employees;
+---------------+
| lowest_salary |
+---------------+
|      35000.00 |
+---------------+
1 row in set (0.001 sec)

mysql>
```

Figure 8: 13 sql query 07 min salary

Source file: 13_sql_query_07_min_salary.png

SCREENSHOT EVIDENCE 09 OF 58

```
mysql> SELECT * FROM employees
    -> ORDER BY salary DESC;
+-------------+--------+------------+----------+-----------+
| employee_id | name   | department | salary   | city      |
+-------------+--------+------------+----------+-----------+
|         107 | Arjun  | Finance    | 55000.00 | Bangalore |
|         114 | Pooja  | Finance    | 53000.00 | Hubli     |
|         103 | Rohan  | Finance    | 52000.00 | Hubli     |
|         108 | Kavya  | IT         | 50000.00 | Mysore    |
|         115 | Aditya | IT         | 49000.00 | Hassan    |
|         104 | Ananya | IT         | 48000.00 | Bangalore |
|         110 | Sneha  | Finance    | 47000.00 | Hassan    |
|         111 | Kiran  | IT         | 46000.00 | Mangalore |
|         101 | Aarav  | IT         | 45000.00 | Bangalore |
|         106 | Meera  | HR         | 42000.00 | Hassan    |
|         112 | Neha   | HR         | 40000.00 | Bangalore |
|         109 | Rahul  | Sales      | 39000.00 | Hubli     |
|         102 | Diya   | HR         | 38000.00 | Mysore    |
|         113 | Manoj  | Sales      | 36000.00 | Mysore    |
|         105 | Vikram | Sales      | 35000.00 | Mangalore |
+-------------+--------+------------+----------+-----------+
15 rows in set (0.002 sec)

mysql>
mysql> SELECT MAX(salary) AS highest_salary
    -> FROM employees;
+----------------+
| highest_salary |
+----------------+
|       55000.00 |
+----------------+
1 row in set (0.003 sec)

mysql>
mysql> SELECT MAX(salary) AS highest_salary
    -> FROM employees;
+----------------+
| highest_salary |
+----------------+
|       55000.00 |
+----------------+
1 row in set (0.001 sec)

mysql>
mysql> SELECT MIN(salary) AS lowest_salary
    -> FROM employees;
+---------------+
| lowest_salary |
+---------------+
|      35000.00 |
+---------------+
1 row in set (0.001 sec)

mysql>
mysql> SELECT AVG(salary) AS average_salary
    -> FROM employees;
+----------------+
| average_salary |
+----------------+
|   45000.000000 |
+----------------+
1 row in set (0.002 sec)

mysql>
```

Figure 9: 14 sql query 08 avg salary

Source file: 14_sql_query_08_avg_salary.png

SCREENSHOT EVIDENCE 10 OF 58

```
|         106 | Meera  | HR          | 42000.00 | Hassan    |
|         112 | Neha   | HR          | 40000.00 | Bangalore |
|         109 | Rahul  | Sales       | 39000.00 | Hubli     |
|         102 | Diya   | HR          | 38000.00 | Mysore    |
|         113 | Manoj  | Sales       | 36000.00 | Mysore    |
|         105 | Vikram | Sales       | 35000.00 | Mangalore |
+-------------+--------+-------------+----------+-----------+
15 rows in set (0.002 sec)

mysql>
mysql> SELECT MAX(salary) AS highest_salary
    -> FROM employees;
+----------------+
| highest_salary |
+----------------+
|       55000.00 |
+----------------+
1 row in set (0.003 sec)

mysql>
mysql> SELECT MAX(salary) AS highest_salary
    -> FROM employees;
+----------------+
| highest_salary |
+----------------+
|       55000.00 |
+----------------+
1 row in set (0.001 sec)

mysql>
mysql> SELECT MIN(salary) AS lowest_salary
    -> FROM employees;
+---------------+
| lowest_salary |
+---------------+
|      35000.00 |
+---------------+
1 row in set (0.001 sec)

mysql>
mysql> SELECT AVG(salary) AS average_salary
    -> FROM employees;
+----------------+
| average_salary |
+----------------+
|   45000.000000 |
+----------------+
1 row in set (0.002 sec)

mysql>
mysql> SELECT department, COUNT(*) AS employee_count
    -> FROM employees
    -> GROUP BY department;
+-------------+----------------+
| department | employee_count |
+-------------+----------------+
| IT          |              5 |
| HR          |              3 |
| Finance     |              4 |
| Sales       |              3 |
+-------------+----------------+
4 rows in set (0.001 sec)

mysql>
```

Figure 10: 15 sql query 09 group by

Source file: 15_sql_query_09_group_by.png

SCREENSHOT EVIDENCE 11 OF 58

```
mysql> SELECT MAX(salary) AS highest_salary
    -> FROM employees;
+----------------+
| highest_salary |
+----------------+
|       55000.00 |
+----------------+
1 row in set (0.003 sec)

mysql>
mysql> SELECT MAX(salary) AS highest_salary
    -> FROM employees;
+----------------+
| highest_salary |
+----------------+
|       55000.00 |
+----------------+
1 row in set (0.001 sec)

mysql>
mysql> SELECT MIN(salary) AS lowest_salary
    -> FROM employees;
+---------------+
| lowest_salary |
+---------------+
|      35000.00 |
+---------------+
1 row in set (0.001 sec)

mysql>
mysql> SELECT AVG(salary) AS average_salary
    -> FROM employees;
+----------------+
| average_salary |
+----------------+
|   45000.000000 |
+----------------+
1 row in set (0.002 sec)

mysql>
mysql> SELECT department, COUNT(*) AS employee_count
    -> FROM employees
    -> GROUP BY department;
+------------+----------------+
| department | employee_count |
+------------+----------------+
| IT         |              5 |
| HR         |              3 |
| Finance    |              4 |
| Sales      |              3 |
+------------+----------------+
4 rows in set (0.001 sec)

mysql>
mysql> SELECT SUM(salary) AS total_salary
    -> FROM employees;
+--------------+
| total_salary |
+--------------+
|    675000.00 |
+--------------+
1 row in set (0.003 sec)

mysql>
```

Figure 11: 16 sql query 10 sum salary

Source file: 16_sql_query_10_sum_salary.png

SCREENSHOT EVIDENCE 12 OF 58

```
| highest_salary |
+----------------+
|       55000.00 |
+----------------+
1 row in set (0.001 sec)

mysql>
mysql> SELECT MIN(salary) AS lowest_salary
    -> FROM employees;
+---------------+
| lowest_salary |
+---------------+
|      35000.00 |
+---------------+
1 row in set (0.001 sec)

mysql>
mysql> SELECT AVG(salary) AS average_salary
    -> FROM employees;
+----------------+
| average_salary |
+----------------+
|   45000.000000 |
+----------------+
1 row in set (0.002 sec)

mysql>
mysql> SELECT department, COUNT(*) AS employee_count
    -> FROM employees
    -> GROUP BY department;
+------------+----------------+
| department | employee_count |
+------------+----------------+
| IT         |              5 |
| HR         |              3 |
| Finance    |              4 |
| Sales      |              3 |
+------------+----------------+
4 rows in set (0.001 sec)

mysql>
mysql> SELECT SUM(salary) AS total_salary
    -> FROM employees;
+--------------+
| total_salary |
+--------------+
|    675000.00 |
+--------------+
1 row in set (0.003 sec)

mysql>
mysql> SELECT * FROM employees
    -> WHERE city = 'Bangalore';
+-------------+---------+------------+----------+-----------+
| employee_id | name    | department | salary   | city      |
+-------------+---------+------------+----------+-----------+
|         101 | Aarav   | IT         | 45000.00 | Bangalore |
|         104 | Ananya  | IT         | 48000.00 | Bangalore |
|         107 | Arjun   | Finance    | 55000.00 | Bangalore |
|         112 | Neha    | HR         | 40000.00 | Bangalore |
+-------------+---------+------------+----------+-----------+
4 rows in set (0.001 sec)

mysql>
```

Figure 12: 17 sql query 11 city filter

Source file: 17_sql_query_11_city_filter.png

SCREENSHOT EVIDENCE 13 OF 58

```
mysql> SELECT AVG(salary) AS average_salary
    -> FROM employees;
+----------------+
| average_salary |
+----------------+
|   45000.000000 |
+----------------+
1 row in set (0.002 sec)

mysql>
mysql> SELECT department, COUNT(*) AS employee_count
    -> FROM employees
    -> GROUP BY department;
+------------+----------------+
| department | employee_count |
+------------+----------------+
| IT         |              5 |
| HR         |              3 |
| Finance    |              4 |
| Sales      |              3 |
+------------+----------------+
4 rows in set (0.001 sec)

mysql>
mysql> SELECT SUM(salary) AS total_salary
    -> FROM employees;
+--------------+
| total_salary |
+--------------+
|    675000.00 |
+--------------+
1 row in set (0.003 sec)

mysql>
mysql> SELECT * FROM employees
    -> WHERE city = 'Bangalore';
+-------------+--------+------------+----------+-----------+
| employee_id | name   | department | salary   | city      |
+-------------+--------+------------+----------+-----------+
|         101 | Aarav  | IT         | 45000.00 | Bangalore |
|         104 | Ananya | IT         | 48000.00 | Bangalore |
|         107 | Arjun  | Finance    | 55000.00 | Bangalore |
|         112 | Neha   | HR         | 40000.00 | Bangalore |
+-------------+--------+------------+----------+-----------+
4 rows in set (0.001 sec)

mysql>
mysql> SELECT * FROM employees
    -> WHERE salary BETWEEN 40000 AND 50000;
+-------------+--------+------------+----------+-----------+
| employee_id | name   | department | salary   | city      |
+-------------+--------+------------+----------+-----------+
|         101 | Aarav  | IT         | 45000.00 | Bangalore |
|         104 | Ananya | IT         | 48000.00 | Bangalore |
|         106 | Meera  | HR         | 42000.00 | Hassan    |
|         108 | Kavya  | IT         | 50000.00 | Mysore    |
|         110 | Sneha  | Finance    | 47000.00 | Hassan    |
|         111 | Kiran  | IT         | 46000.00 | Mangalore |
|         112 | Neha   | HR         | 40000.00 | Bangalore |
|         115 | Aditya | IT         | 49000.00 | Hassan    |
+-------------+--------+------------+----------+-----------+
8 rows in set (0.002 sec)

mysql>
```

Figure 13: 18 sql query 12 between salary

Source file: 18_sql_query_12_between_salary.png

SCREENSHOT EVIDENCE 14 OF 58

```
+-------------+----------------+
| department | employee_count |
+-------------+----------------+
| IT         |              5 |
| HR         |              3 |
| Finance    |              4 |
| Sales      |              3 |
+-------------+----------------+
4 rows in set (0.001 sec)

mysql>
mysql> SELECT SUM(salary) AS total_salary
    -> FROM employees;
+--------------+
| total_salary |
+--------------+
|    675000.00 |
+--------------+
1 row in set (0.003 sec)

mysql>
mysql> SELECT * FROM employees
    -> WHERE city = 'Bangalore';
+-------------+--------+------------+----------+-----------+
| employee_id | name   | department | salary   | city      |
+-------------+--------+------------+----------+-----------+
|         101 | Aarav  | IT         | 45000.00 | Bangalore |
|         104 | Ananya | IT         | 48000.00 | Bangalore |
|         107 | Arjun  | Finance    | 55000.00 | Bangalore |
|         112 | Neha   | HR         | 40000.00 | Bangalore |
+-------------+--------+------------+----------+-----------+
4 rows in set (0.001 sec)

mysql>
mysql> SELECT * FROM employees
    -> WHERE salary BETWEEN 40000 AND 50000;
+-------------+--------+------------+----------+-----------+
| employee_id | name   | department | salary   | city      |
+-------------+--------+------------+----------+-----------+
|         101 | Aarav  | IT         | 45000.00 | Bangalore |
|         104 | Ananya | IT         | 48000.00 | Bangalore |
|         106 | Meera  | HR         | 42000.00 | Hassan    |
|         108 | Kavya  | IT         | 50000.00 | Mysore    |
|         110 | Sneha  | Finance    | 47000.00 | Hassan    |
|         111 | Kiran  | IT         | 46000.00 | Mangalore |
|         112 | Neha   | HR         | 40000.00 | Bangalore |
|         115 | Aditya | IT         | 49000.00 | Hassan    |
+-------------+--------+------------+----------+-----------+
8 rows in set (0.002 sec)

mysql>
mysql> SELECT * FROM employees
    -> WHERE name LIKE 'A%';
+-------------+--------+------------+----------+-----------+
| employee_id | name   | department | salary   | city      |
+-------------+--------+------------+----------+-----------+
|         101 | Aarav  | IT         | 45000.00 | Bangalore |
|         104 | Ananya | IT         | 48000.00 | Bangalore |
|         107 | Arjun  | Finance    | 55000.00 | Bangalore |
|         115 | Aditya | IT         | 49000.00 | Hassan    |
+-------------+--------+------------+----------+-----------+
4 rows in set (0.002 sec)

mysql>
```

Figure 14: 19 sql query 13 like names

Source file: 19_sql_query_13_like_names.png

SCREENSHOT EVIDENCE 15 OF 58

```
+---------------+
| total_salary |
+---------------+
|    675000.00 |
+---------------+
1 row in set (0.003 sec)

mysql>
mysql> SELECT * FROM employees
    -> WHERE city = 'Bangalore';
+-------------+--------+------------+----------+-----------+
| employee_id | name   | department | salary   | city      |
+-------------+--------+------------+----------+-----------+
|         101 | Aarav  | IT         | 45000.00 | Bangalore |
|         104 | Ananya | IT         | 48000.00 | Bangalore |
|         107 | Arjun  | Finance    | 55000.00 | Bangalore |
|         112 | Neha   | HR         | 40000.00 | Bangalore |
+-------------+--------+------------+----------+-----------+
4 rows in set (0.001 sec)

mysql>
mysql> SELECT * FROM employees
    -> WHERE salary BETWEEN 40000 AND 50000;
+-------------+--------+------------+----------+-----------+
| employee_id | name   | department | salary   | city      |
+-------------+--------+------------+----------+-----------+
|         101 | Aarav  | IT         | 45000.00 | Bangalore |
|         104 | Ananya | IT         | 48000.00 | Bangalore |
|         106 | Meera  | HR         | 42000.00 | Hassan    |
|         108 | Kavya  | IT         | 50000.00 | Mysore    |
|         110 | Sneha  | Finance    | 47000.00 | Hassan    |
|         111 | Kiran  | IT         | 46000.00 | Mangalore |
|         112 | Neha   | HR         | 40000.00 | Bangalore |
|         115 | Aditya | IT         | 49000.00 | Hassan    |
+-------------+--------+------------+----------+-----------+
8 rows in set (0.002 sec)

mysql>
mysql> SELECT * FROM employees
    -> WHERE name LIKE 'A%';
+-------------+--------+------------+----------+-----------+
| employee_id | name   | department | salary   | city      |
+-------------+--------+------------+----------+-----------+
|         101 | Aarav  | IT         | 45000.00 | Bangalore |
|         104 | Ananya | IT         | 48000.00 | Bangalore |
|         107 | Arjun  | Finance    | 55000.00 | Bangalore |
|         115 | Aditya | IT         | 49000.00 | Hassan    |
+-------------+--------+------------+----------+-----------+
4 rows in set (0.002 sec)

mysql>
mysql> SELECT * FROM employees
    -> WHERE salary >= 50000;
+-------------+-------+------------+----------+-----------+
| employee_id | name  | department | salary   | city      |
+-------------+-------+------------+----------+-----------+
|         103 | Rohan | Finance    | 52000.00 | Hubli     |
|         107 | Arjun | Finance    | 55000.00 | Bangalore |
|         108 | Kavya | IT         | 50000.00 | Mysore    |
|         114 | Pooja | Finance    | 53000.00 | Hubli     |
+-------------+-------+------------+----------+-----------+
4 rows in set (0.001 sec)

mysql>
```

Figure 15: 20 sql query 14 salary filter

Source file: 20_sql_query_14_salary_filter.png

SCREENSHOT EVIDENCE 16 OF 58

```
|         107 | Arjun   | Finance    | 55000.00 | Bangalore |
|         112 | Neha    | HR         | 40000.00 | Bangalore |
+-------------+---------+------------+----------+-----------+
4 rows in set (0.001 sec)

mysql>
mysql> SELECT * FROM employees
    -> WHERE salary BETWEEN 40000 AND 50000;
+-------------+---------+------------+----------+-----------+
| employee_id | name    | department | salary   | city      |
+-------------+---------+------------+----------+-----------+
|         101 | Aarav   | IT         | 45000.00 | Bangalore |
|         104 | Ananya  | IT         | 48000.00 | Bangalore |
|         106 | Meera   | HR         | 42000.00 | Hassan    |
|         108 | Kavya   | IT         | 50000.00 | Mysore    |
|         110 | Sneha   | Finance    | 47000.00 | Hassan    |
|         111 | Kiran   | IT         | 46000.00 | Mangalore |
|         112 | Neha    | HR         | 40000.00 | Bangalore |
|         115 | Aditya  | IT         | 49000.00 | Hassan    |
+-------------+---------+------------+----------+-----------+
8 rows in set (0.002 sec)

mysql>
mysql> SELECT * FROM employees
    -> WHERE name LIKE 'A%';
+-------------+---------+------------+----------+-----------+
| employee_id | name    | department | salary   | city      |
+-------------+---------+------------+----------+-----------+
|         101 | Aarav   | IT         | 45000.00 | Bangalore |
|         104 | Ananya  | IT         | 48000.00 | Bangalore |
|         107 | Arjun   | Finance    | 55000.00 | Bangalore |
|         115 | Aditya  | IT         | 49000.00 | Hassan    |
+-------------+---------+------------+----------+-----------+
4 rows in set (0.002 sec)

mysql>
mysql> SELECT * FROM employees
    -> WHERE salary >= 50000;
+-------------+---------+------------+----------+-----------+
| employee_id | name    | department | salary   | city      |
+-------------+---------+------------+----------+-----------+
|         103 | Rohan   | Finance    | 52000.00 | Hubli     |
|         107 | Arjun   | Finance    | 55000.00 | Bangalore |
|         108 | Kavya   | IT         | 50000.00 | Mysore    |
|         114 | Pooja   | Finance    | 53000.00 | Hubli     |
+-------------+---------+------------+----------+-----------+
4 rows in set (0.001 sec)

mysql>
mysql> SELECT * FROM employees
    -> WHERE city IN ('Mysore', 'Hassan');
+-------------+---------+------------+----------+---------+
| employee_id | name    | department | salary   | city    |
+-------------+---------+------------+----------+---------+
|         102 | Diya    | HR         | 38000.00 | Mysore  |
|         106 | Meera   | HR         | 42000.00 | Hassan  |
|         108 | Kavya   | IT         | 50000.00 | Mysore  |
|         110 | Sneha   | Finance    | 47000.00 | Hassan  |
|         113 | Manoj   | Sales      | 36000.00 | Mysore  |
|         115 | Aditya  | IT         | 49000.00 | Hassan  |
+-------------+---------+------------+----------+---------+
6 rows in set (0.001 sec)

mysql>
```

Figure 16: 21 sql query 15 in cities

Source file: 21_sql_query_15_in_cities.png

SCREENSHOT EVIDENCE 17 OF 58

```
|         108 | Kavya  | IT         | 50000.00 | Mysore    |
|         110 | Sneha  | Finance    | 47000.00 | Hassan    |
|         111 | Kiran  | IT         | 46000.00 | Mangalore |
|         112 | Neha   | HR         | 40000.00 | Bangalore |
|         115 | Aditya | IT         | 49000.00 | Hassan    |
+-------------+--------+------------+----------+-----------+
8 rows in set (0.002 sec)

mysql>
mysql> SELECT * FROM employees
    -> WHERE name LIKE 'A%';
+-------------+--------+------------+----------+-----------+
| employee_id | name   | department | salary   | city      |
+-------------+--------+------------+----------+-----------+
|         101 | Aarav  | IT         | 45000.00 | Bangalore |
|         104 | Ananya | IT         | 48000.00 | Bangalore |
|         107 | Arjun  | Finance    | 55000.00 | Bangalore |
|         115 | Aditya | IT         | 49000.00 | Hassan    |
+-------------+--------+------------+----------+-----------+
4 rows in set (0.002 sec)

mysql>
mysql> SELECT * FROM employees
    -> WHERE salary >= 50000;
+-------------+--------+------------+----------+-----------+
| employee_id | name   | department | salary   | city      |
+-------------+--------+------------+----------+-----------+
|         103 | Rohan  | Finance    | 52000.00 | Hubli     |
|         107 | Arjun  | Finance    | 55000.00 | Bangalore |
|         108 | Kavya  | IT         | 50000.00 | Mysore    |
|         114 | Pooja  | Finance    | 53000.00 | Hubli     |
+-------------+--------+------------+----------+-----------+
4 rows in set (0.001 sec)

mysql>
mysql> SELECT * FROM employees
    -> WHERE city IN ('Mysore', 'Hassan');
+-------------+--------+------------+----------+---------+
| employee_id | name   | department | salary   | city    |
+-------------+--------+------------+----------+---------+
|         102 | Diya   | HR         | 38000.00 | Mysore  |
|         106 | Meera  | HR         | 42000.00 | Hassan  |
|         108 | Kavya  | IT         | 50000.00 | Mysore  |
|         110 | Sneha  | Finance    | 47000.00 | Hassan  |
|         113 | Manoj  | Sales      | 36000.00 | Mysore  |
|         115 | Aditya | IT         | 49000.00 | Hassan  |
+-------------+--------+------------+----------+---------+
6 rows in set (0.001 sec)

mysql>
mysql> SELECT * FROM employees
    -> WHERE department = 'IT'
    -> AND salary > 45000;
+-------------+--------+------------+----------+-----------+
| employee_id | name   | department | salary   | city      |
+-------------+--------+------------+----------+-----------+
|         104 | Ananya | IT         | 48000.00 | Bangalore |
|         108 | Kavya  | IT         | 50000.00 | Mysore    |
|         111 | Kiran  | IT         | 46000.00 | Mangalore |
|         115 | Aditya | IT         | 49000.00 | Hassan    |
+-------------+--------+------------+----------+-----------+
4 rows in set (0.000 sec)

mysql>
```

Figure 17: 22 sql query 16 and condition

Source file: 22_sql_query_16_and_condition.png

SCREENSHOT EVIDENCE 18 OF 58

```
|         107 | Arjun   | Finance    | 55000.00 | Bangalore |
|         115 | Aditya  | IT         | 49000.00 | Hassan    |
+-------------+---------+------------+----------+-----------+
4 rows in set (0.002 sec)

mysql>
mysql> SELECT * FROM employees
    -> WHERE salary >= 50000;
+-------------+---------+------------+----------+-----------+
| employee_id | name    | department | salary   | city      |
+-------------+---------+------------+----------+-----------+
|         103 | Rohan   | Finance    | 52000.00 | Hubli     |
|         107 | Arjun   | Finance    | 55000.00 | Bangalore |
|         108 | Kavya   | IT         | 50000.00 | Mysore    |
|         114 | Pooja   | Finance    | 53000.00 | Hubli     |
+-------------+---------+------------+----------+-----------+
4 rows in set (0.001 sec)

mysql>
mysql> SELECT * FROM employees
    -> WHERE city IN ('Mysore', 'Hassan');
+-------------+---------+------------+----------+--------+
| employee_id | name    | department | salary   | city   |
+-------------+---------+------------+----------+--------+
|         102 | Diya    | HR         | 38000.00 | Mysore |
|         106 | Meera   | HR         | 42000.00 | Hassan |
|         108 | Kavya   | IT         | 50000.00 | Mysore |
|         110 | Sneha   | Finance    | 47000.00 | Hassan |
|         113 | Manoj   | Sales      | 36000.00 | Mysore |
|         115 | Aditya  | IT         | 49000.00 | Hassan |
+-------------+---------+------------+----------+--------+
6 rows in set (0.001 sec)

mysql>
mysql> SELECT * FROM employees
    -> WHERE department = 'IT'
    -> AND salary > 45000;
+-------------+---------+------------+----------+-----------+
| employee_id | name    | department | salary   | city      |
+-------------+---------+------------+----------+-----------+
|         104 | Ananya  | IT         | 48000.00 | Bangalore |
|         108 | Kavya   | IT         | 50000.00 | Mysore    |
|         111 | Kiran   | IT         | 46000.00 | Mangalore |
|         115 | Aditya  | IT         | 49000.00 | Hassan    |
+-------------+---------+------------+----------+-----------+
4 rows in set (0.000 sec)

mysql>
mysql> SELECT * FROM employees
    -> WHERE department = 'HR'
    -> OR department = 'Sales';
+-------------+---------+------------+----------+-----------+
| employee_id | name    | department | salary   | city      |
+-------------+---------+------------+----------+-----------+
|         102 | Diya    | HR         | 38000.00 | Mysore    |
|         105 | Vikram  | Sales      | 35000.00 | Mangalore |
|         106 | Meera   | HR         | 42000.00 | Hassan    |
|         109 | Rahul   | Sales      | 39000.00 | Hubli     |
|         112 | Neha    | HR         | 40000.00 | Bangalore |
|         113 | Manoj   | Sales      | 36000.00 | Mysore    |
+-------------+---------+------------+----------+-----------+
6 rows in set (0.001 sec)

mysql>
```

Figure 18: 23 sql query 17 or condition

Source file: 23_sql_query_17_or_condition.png

SCREENSHOT EVIDENCE 19 OF 58

```
    -> OR department = 'Sales';
+-------------+---------+------------+----------+-----------+
| employee_id | name    | department | salary   | city      |
+-------------+---------+------------+----------+-----------+
|         102 | Diya    | HR         | 38000.00 | Mysore    |
|         105 | Vikram  | Sales      | 35000.00 | Mangalore |
|         106 | Meera   | HR         | 42000.00 | Hassan    |
|         109 | Rahul   | Sales      | 39000.00 | Hubli     |
|         112 | Neha    | HR         | 40000.00 | Bangalore |
|         113 | Manoj   | Sales      | 36000.00 | Mysore    |
+-------------+---------+------------+----------+-----------+
6 rows in set (0.001 sec)

mysql>
mysql> SELECT * FROM employees
    -> ORDER BY name ASC;
+-------------+---------+------------+----------+-----------+
| employee_id | name    | department | salary   | city      |
+-------------+---------+------------+----------+-----------+
|         101 | Aarav   | IT         | 45000.00 | Bangalore |
|         115 | Aditya  | IT         | 49000.00 | Hassan    |
|         104 | Ananya  | IT         | 48000.00 | Bangalore |
|         107 | Arjun   | Finance    | 55000.00 | Bangalore |
|         102 | Diya    | HR         | 38000.00 | Mysore    |
|         108 | Kavya   | IT         | 50000.00 | Mysore    |
|         111 | Kiran   | IT         | 46000.00 | Mangalore |
|         113 | Manoj   | Sales      | 36000.00 | Mysore    |
|         106 | Meera   | HR         | 42000.00 | Hassan    |
|         112 | Neha    | HR         | 40000.00 | Bangalore |
|         114 | Pooja   | Finance    | 53000.00 | Hubli     |
|         109 | Rahul   | Sales      | 39000.00 | Hubli     |
|         103 | Rohan   | Finance    | 52000.00 | Hubli     |
|         110 | Sneha   | Finance    | 47000.00 | Hassan    |
|         105 | Vikram  | Sales      | 35000.00 | Mangalore |
+-------------+---------+------------+----------+-----------+
15 rows in set (0.002 sec)

mysql>
mysql> SELECT city, COUNT(*) AS employee_count
    -> FROM employees
    -> GROUP BY city
    -> ORDER BY city;
+-----------+----------------+
| city      | employee_count |
+-----------+----------------+
| Bangalore |              4 |
| Hassan    |              3 |
| Hubli     |              3 |
| Mangalore |              2 |
| Mysore    |              3 |
+-----------+----------------+
5 rows in set (0.001 sec)

mysql>
mysql> SELECT * FROM employees
    -> WHERE salary = (SELECT MAX(salary) FROM employees);
+-------------+---------+------------+----------+-----------+
| employee_id | name    | department | salary   | city      |
+-------------+---------+------------+----------+-----------+
|         107 | Arjun   | Finance    | 55000.00 | Bangalore |
+-------------+---------+------------+----------+-----------+
1 row in set (0.001 sec)

mysql>
```

Figure 19: 26 sql query 20 highest paid

Source file: 26_sql_query_20_highest_paid.png

SCREENSHOT EVIDENCE 20 OF 58

```
mysql>
mysql> SELECT * FROM employees
    -> ORDER BY name ASC;
+-------------+--------+------------+----------+-----------+
| employee_id | name   | department | salary   | city      |
+-------------+--------+------------+----------+-----------+
|         101 | Aarav  | IT         | 45000.00 | Bangalore |
|         115 | Aditya | IT         | 49000.00 | Hassan    |
|         104 | Ananya | IT         | 48000.00 | Bangalore |
|         107 | Arjun  | Finance    | 55000.00 | Bangalore |
|         102 | Diya   | HR         | 38000.00 | Mysore    |
|         108 | Kavya  | IT         | 50000.00 | Mysore    |
|         111 | Kiran  | IT         | 46000.00 | Mangalore |
|         113 | Manoj  | Sales      | 36000.00 | Mysore    |
|         106 | Meera  | HR         | 42000.00 | Hassan    |
|         112 | Neha   | HR         | 40000.00 | Bangalore |
|         114 | Pooja  | Finance    | 53000.00 | Hubli     |
|         109 | Rahul  | Sales      | 39000.00 | Hubli     |
|         103 | Rohan  | Finance    | 52000.00 | Hubli     |
|         110 | Sneha  | Finance    | 47000.00 | Hassan    |
|         105 | Vikram | Sales      | 35000.00 | Mangalore |
+-------------+--------+------------+----------+-----------+
15 rows in set (0.002 sec)

mysql>
mysql> SELECT city, COUNT(*) AS employee_count
    -> FROM employees
    -> GROUP BY city
    -> ORDER BY city;
+-----------+----------------+
| city      | employee_count |
+-----------+----------------+
| Bangalore |              4 |
| Hassan    |              3 |
| Hubli     |              3 |
| Mangalore |              2 |
| Mysore    |              3 |
+-----------+----------------+
5 rows in set (0.001 sec)

mysql>
mysql> SELECT * FROM employees
    -> WHERE salary = (SELECT MAX(salary) FROM employees);
+-------------+------+------------+----------+-----------+
| employee_id | name | department | salary   | city      |
+-------------+------+------------+----------+-----------+
|         107 | Arjun | Finance   | 55000.00 | Bangalore |
+-------------+------+------------+----------+-----------+
1 row in set (0.001 sec)

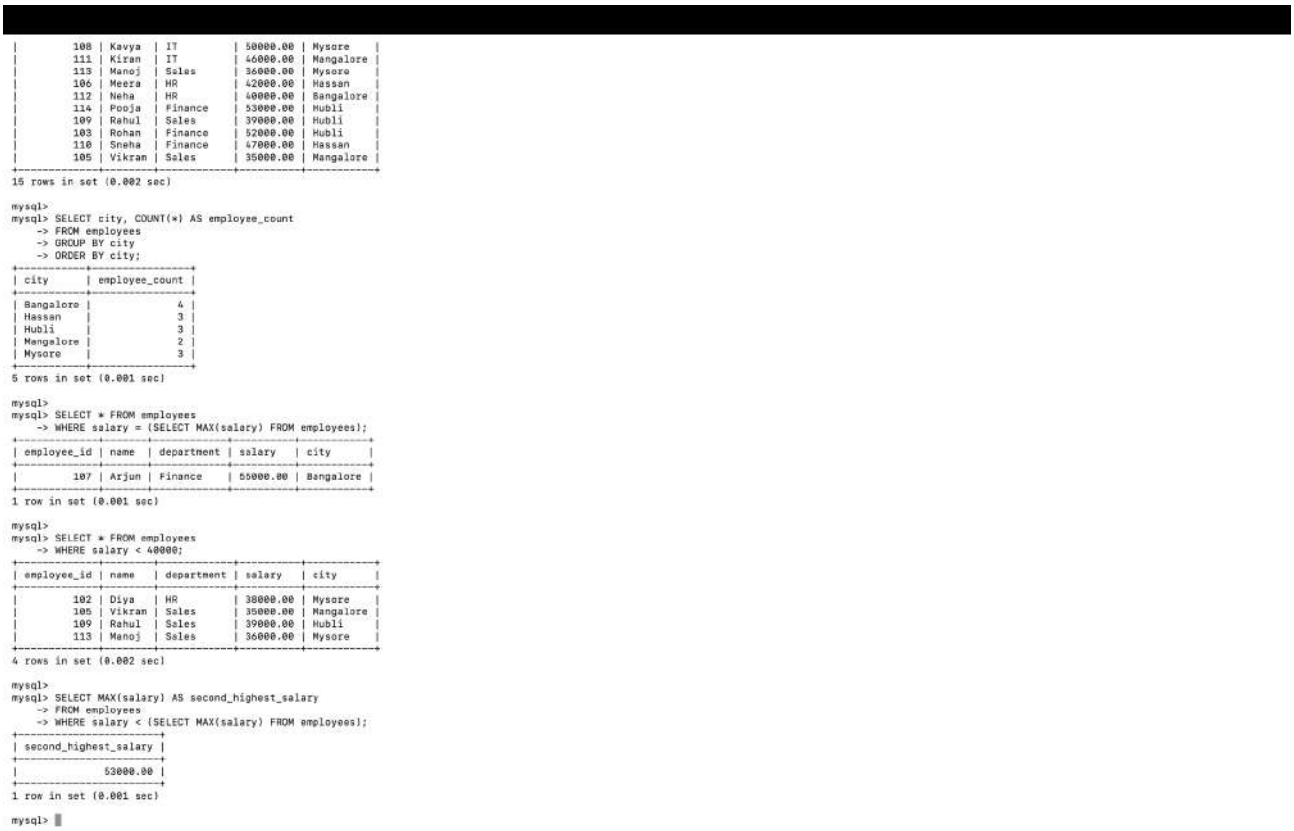
mysql>
mysql> SELECT * FROM employees
    -> WHERE salary < 40000;
+-------------+--------+------------+----------+-----------+
| employee_id | name   | department | salary   | city      |
+-------------+--------+------------+----------+-----------+
|         102 | Diya   | HR         | 38000.00 | Mysore    |
|         105 | Vikram | Sales      | 35000.00 | Mangalore |
|         109 | Rahul  | Sales      | 39000.00 | Hubli     |
|         113 | Manoj  | Sales      | 36000.00 | Mysore    |
+-------------+--------+------------+----------+-----------+
4 rows in set (0.002 sec)

mysql>
```

Figure 20: 27 sql query 21 below salary

Source file: 27_sql_query_21_below_salary.png

SCREENSHOT EVIDENCE 21 OF 58



```
|         108 | Kavya  | IT          | 50000.00 | Mysore    |
|         111 | Kiran  | IT          | 46000.00 | Mangalore |
|         113 | Manoj  | Sales       | 36000.00 | Mysore    |
|         106 | Meera  | HR          | 42000.00 | Hassan    |
|         112 | Neha   | HR          | 40000.00 | Bangalore |
|         114 | Pooja  | Finance     | 53000.00 | Hubli     |
|         109 | Rahul  | Sales       | 39000.00 | Hubli     |
|         103 | Rohan  | Finance     | 52000.00 | Hubli     |
|         110 | Sneha  | Finance     | 47000.00 | Hassan    |
|         105 | Vikran | Sales       | 35000.00 | Mangalore |
+-------------+--------+-------------+----------+-----------+
15 rows in set (0.002 sec)

mysql>
mysql> SELECT city, COUNT(*) AS employee_count
    -> FROM employees
    -> GROUP BY city
    -> ORDER BY city;
+-----------+----------------+
| city      | employee_count |
+-----------+----------------+
| Bangalore |              4 |
| Hassan    |              3 |
| Hubli     |              3 |
| Mangalore |              2 |
| Mysore    |              3 |
+-----------+----------------+
5 rows in set (0.001 sec)

mysql>
mysql> SELECT * FROM employees
    -> WHERE salary = (SELECT MAX(salary) FROM employees);
+-------------+------+------------+----------+-----------+
| employee_id | name | department | salary   | city      |
+-------------+------+------------+----------+-----------+
|         107 | Arjun | Finance    | 55000.00 | Bangalore |
+-------------+------+------------+----------+-----------+
1 row in set (0.001 sec)

mysql>
mysql> SELECT * FROM employees
    -> WHERE salary < 40000;
+-------------+--------+------------+----------+-----------+
| employee_id | name   | department | salary   | city      |
+-------------+--------+------------+----------+-----------+
|         102 | Diya   | HR         | 38000.00 | Mysore    |
|         105 | Vikran | Sales      | 35000.00 | Mangalore |
|         109 | Rahul  | Sales      | 39000.00 | Hubli     |
|         113 | Manoj  | Sales      | 36000.00 | Mysore    |
+-------------+--------+------------+----------+-----------+
4 rows in set (0.002 sec)

mysql>
mysql> SELECT MAX(salary) AS second_highest_salary
    -> FROM employees
    -> WHERE salary < (SELECT MAX(salary) FROM employees);
+-----------------------+
| second_highest_salary |
+-----------------------+
|              53000.00 |
+-----------------------+
1 row in set (0.001 sec)

mysql>
```

Figure 21: 28 sql query 22 second highest salary

Source file: 28_sql_query_22_second_highest_salary.png

SCREENSHOT EVIDENCE 22 OF 58

```
    -> WHERE salary = (SELECT MAX(salary) FROM employees);
+-------------+--------+------------+----------+-----------+
| employee_id | name   | department | salary   | city      |
+-------------+--------+------------+----------+-----------+
|         107 | Arjun  | Finance    | 55000.00 | Bangalore |
+-------------+--------+------------+----------+-----------+
1 row in set (0.001 sec)

mysql>
mysql> SELECT * FROM employees
    -> WHERE salary < 40000;
+-------------+--------+------------+----------+-----------+
| employee_id | name   | department | salary   | city      |
+-------------+--------+------------+----------+-----------+
|         102 | Diya   | HR         | 38000.00 | Mysore    |
|         105 | Vikram | Sales      | 35000.00 | Mangalore |
|         109 | Rahul  | Sales      | 39000.00 | Hubli     |
|         113 | Manoj  | Sales      | 36000.00 | Mysore    |
+-------------+--------+------------+----------+-----------+
4 rows in set (0.002 sec)

mysql>
mysql> SELECT MAX(salary) AS second_highest_salary
    -> FROM employees
    -> WHERE salary < (SELECT MAX(salary) FROM employees);
+------------------------+
| second_highest_salary  |
+------------------------+
|              53000.00  |
+------------------------+
1 row in set (0.001 sec)

mysql>
mysql> SELECT *
    -> FROM employees
    -> WHERE salary > (SELECT AVG(salary) FROM employees);
+-------------+--------+------------+----------+-----------+
| employee_id | name   | department | salary   | city      |
+-------------+--------+------------+----------+-----------+
|         103 | Rohan  | Finance    | 52000.00 | Hubli     |
|         104 | Ananya | IT         | 48000.00 | Bangalore |
|         107 | Arjun  | Finance    | 55000.00 | Bangalore |
|         108 | Kavya  | IT         | 50000.00 | Mysore    |
|         110 | Sneha  | Finance    | 47000.00 | Hassan    |
|         111 | Kiran  | IT         | 46000.00 | Mangalore |
|         114 | Pooja  | Finance    | 53000.00 | Hubli     |
|         115 | Aditya | IT         | 49000.00 | Hassan    |
+-------------+--------+------------+----------+-----------+
8 rows in set (0.001 sec)

mysql>
mysql> SELECT department, COUNT(*) AS employee_count
    -> FROM employees
    -> WHERE salary > 45000
    -> GROUP BY department;
+------------+----------------+
| department | employee_count |
+------------+----------------+
| Finance    |              4 |
| IT         |              4 |
+------------+----------------+
2 rows in set (0.001 sec)

mysql>
```

Figure 22: 30 sql query 24 department salary count

Source file: 30_sql_query_24_department_salary_count.png

SCREENSHOT EVIDENCE 23 OF 58

```
+-------------+---------+-------------+-----------+------------+
4 rows in set (0.002 sec)

mysql>
mysql> SELECT MAX(salary) AS second_highest_salary
    -> FROM employees
    -> WHERE salary < (SELECT MAX(salary) FROM employees);
+------------------------+
| second_highest_salary |
+------------------------+
|              53000.00 |
+------------------------+
1 row in set (0.001 sec)

mysql>
mysql> SELECT *
    -> FROM employees
    -> WHERE salary > (SELECT AVG(salary) FROM employees);
+-------------+---------+-------------+-----------+------------+
| employee_id | name    | department | salary    | city       |
+-------------+---------+-------------+-----------+------------+
|         103 | Rohan   | Finance     | 52000.00 | Hubli      |
|         104 | Ananya  | IT          | 48000.00 | Bangalore  |
|         107 | Arjun   | Finance     | 55000.00 | Bangalore  |
|         108 | Kavya   | IT          | 50000.00 | Mysore     |
|         110 | Sneha   | Finance     | 47000.00 | Hassan     |
|         111 | Kiran   | IT          | 46000.00 | Mangalore  |
|         114 | Pooja   | Finance     | 53000.00 | Hubli      |
|         115 | Aditya  | IT          | 49000.00 | Hassan     |
+-------------+---------+-------------+-----------+------------+
8 rows in set (0.001 sec)

mysql>
mysql> SELECT department, COUNT(*) AS employee_count
    -> FROM employees
    -> WHERE salary > 45000
    -> GROUP BY department;
+------------+----------------+
| department | employee_count |
+------------+----------------+
| Finance    |              4 |
| IT         |              4 |
+------------+----------------+
2 rows in set (0.001 sec)

mysql>
mysql> SELECT *
    -> FROM employees e
    -> WHERE salary = (
    ->     SELECT MIN(salary)
    ->     FROM employees
    ->     WHERE department = e.department
    -> );
+-------------+---------+-------------+-----------+------------+
| employee_id | name    | department | salary    | city       |
+-------------+---------+-------------+-----------+------------+
|         101 | Aarav   | IT          | 45000.00 | Bangalore  |
|         102 | Diya    | HR          | 38000.00 | Mysore     |
|         105 | Vikram  | Sales       | 35000.00 | Mangalore  |
|         110 | Sneha   | Finance     | 47000.00 | Hassan     |
+-------------+---------+-------------+-----------+------------+
4 rows in set (0.001 sec)

mysql>
```

Figure 23: 31 sql query 25 min salary by department

Source file: 31_sql_query_25_min_salary_by_department.png

SCREENSHOT EVIDENCE 24 OF 58

```
Last login: Sat Oct 10 22:53:50 on ttys001
nandithagn@Nandithas-MacBook-Air ~ % pwd
/Users/nandithagn
nandithagn@Nandithas-MacBook-Air ~ % ls
26BTRCF037(RAKSHITH R).pdf                  DEEKSHITHPROMT.docx                     Library                         Postman
Applications                                Design and thinking_gowtham.pdf         Movies                          Public
APPLIED PHYSICS (RAKSHITH).pdf              Desktop                                 Music                           RAKSHITH_R(26BTRCF037).pdf
AWS_Cloud_Data_Engineering                  Documents                               NANDITHA PROMT.docx             RAKSHITH(Applied Pysics).pdf
cirtificates Nanditha                       Downloads                               PCL_FInal_Paper.pdf             WhatsApp Image 2026-09-21 at 14.25.48.jpeg
cTutorial                                   hcl-data-engineering                    Pictures
nandithagn@Nandithas-MacBook-Air ~ % ls -l
total 24272
-rw-r--r--@    1 nandithagn  staff   919386 22 Sep 19:31 26BTRCF037(RAKSHITH R).pdf
drwx------@    5 nandithagn  staff      160  8 Sep 11:03 Applications
-rw-r--r--@    1 nandithagn  staff   778660 22 Sep 19:11 APPLIED PHYSICS (RAKSHITH).pdf
drwxr-xr-x     8 nandithagn  staff      256 10 Oct 22:59 AWS_Cloud_Data_Engineering
drwxr-xr-x     3 nandithagn  staff       96 12 Apr 19:05 cirtificates Nanditha
drwxr-xr-x     3 nandithagn  staff       96 20 Jul 13:38 cTutorial
-rw-r--r--@    1 nandithagn  staff    14622  2 Sep 11:34 DEEKSHITHPROMT.docx
-rw-r--r--@    1 nandithagn  staff  6726541 23 Sep 23:44 Design and thinking_gowtham.pdf
drwx------+  421 nandithagn  staff    13472 10 Oct 22:57 Desktop
drwx------+    8 nandithagn  staff      256  6 Jul 17:36 Documents
drwx------+ 1096 nandithagn  staff    35072 10 Oct 22:47 Downloads
drwxr-xr-x     8 nandithagn  staff      256  4 Oct 21:48 hcl-data-engineering
drwx------@   96 nandithagn  staff     3072  6 Dec  2025 Library
drwx------     7 nandithagn  staff      224  9 Apr  2026 Movies
drwx------+    6 nandithagn  staff      192 13 Sep  2025 Music
-rw-r--r--@    1 nandithagn  staff    14616  2 Sep 11:33 NANDITHA PROMT.docx
-rw-r--r--@    1 nandithagn  staff  2089356 29 Apr 22:46 PCL_FInal_Paper.pdf
drwx------+    6 nandithagn  staff      192 12 Aug 10:13 Pictures
drwxr-xr-x@    3 nandithagn  staff       96  3 Oct 22:23 Postman
drwxr-xr-x+    4 nandithagn  staff      128 22 Jul  2025 Public
-rw-r--r--@    1 nandithagn  staff   919432 22 Sep 19:34 RAKSHITH_R(26BTRCF037).pdf
-rw-r--r--@    1 nandithagn  staff   921871 22 Sep 19:28 RAKSHITH(Applied Pysics).pdf
-rw-r--r--@    1 nandithagn  staff   186078 21 Sep 14:38 WhatsApp Image 2026-09-21 at 14.25.48.jpeg
nandithagn@Nandithas-MacBook-Air ~ % date
Sun Oct 11 00:10:33 IST 2026
nandithagn@Nandithas-MacBook-Air ~ % whoami
nandithagn
nandithagn@Nandithas-MacBook-Air ~ %
```

Figure 24: 32 linux commands 01 to 05.png.

Source file: 32_linux_commands_01_to_05.png..png

SCREENSHOT EVIDENCE 25 OF 58

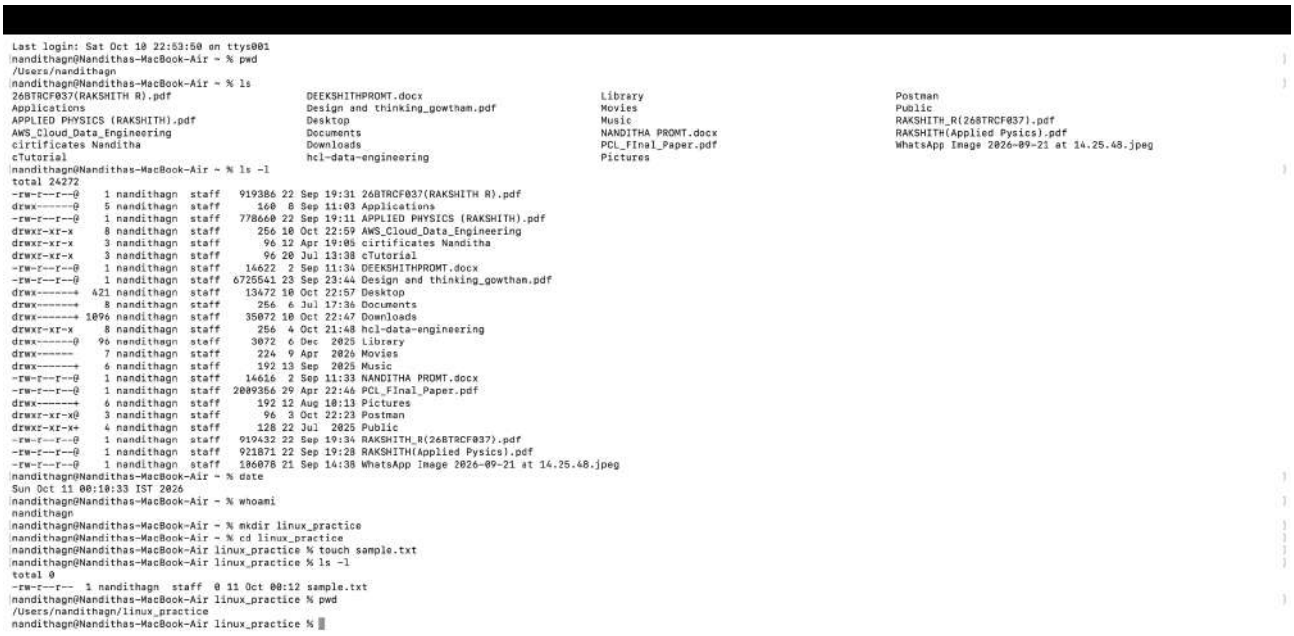


Figure 25: 33 linux commands 06 to 10

Source file: 33_linux_commands_06_to_10.png

SCREENSHOT EVIDENCE 26 OF 58

```
Last login: Sat Oct 10 22:53:50 on ttys001
nandithagn@Nandithas-MacBook-Air ~ % pwd
/Users/nandithagn
nandithagn@Nandithas-MacBook-Air ~ % ls
26BTRCF037(RAKSHITH R).pdf                  DEEKSHITHPROMT.docx                     Library                         Postman
Applications                                Design and thinking_gowtham.pdf         Movies                          Public
APPLIED PHYSICS (RAKSHITH).pdf              Desktop                                 Music                           RAKSHITH_R(26BTRCF037).pdf
AWS_Cloud_Data_Engineering                  Documents                               NANDITHA PROMT.docx             RAKSHITH(Applied Pysics).pdf
cirtificates Nanditha                       Downloads                               PCL_FInal_Paper.pdf             WhatsApp Image 2026-09-21 at 14.25.48.jpeg
cTutorial                                   hcl-data-engineering                    Pictures
nandithagn@Nandithas-MacBook-Air ~ % ls -l
total 24272
-rw-r--r--@    1 nandithagn  staff   919386 22 Sep 19:31 26BTRCF037(RAKSHITH R).pdf
drwx------@    5 nandithagn  staff      160  8 Sep 11:03 Applications
-rw-r--r--@    1 nandithagn  staff   778660 22 Sep 19:11 APPLIED PHYSICS (RAKSHITH).pdf
drwxr-xr-x     8 nandithagn  staff      256 10 Oct 22:59 AWS_Cloud_Data_Engineering
drwxr-xr-x     3 nandithagn  staff       96 12 Apr 19:05 cirtificates Nanditha
drwxr-xr-x     3 nandithagn  staff       96 20 Jul 13:38 cTutorial
-rw-r--r--@    1 nandithagn  staff    14622  2 Sep 11:34 DEEKSHITHPROMT.docx
-rw-r--r--@    1 nandithagn  staff  6726541 23 Sep 23:44 Design and thinking_gowtham.pdf
drwx------+  421 nandithagn  staff    13472 10 Oct 22:57 Desktop
drwx------+    8 nandithagn  staff      256  6 Jul 17:36 Documents
drwx------+ 1096 nandithagn  staff    35072 10 Oct 22:47 Downloads
drwxr-xr-x     8 nandithagn  staff      256  4 Oct 21:48 hcl-data-engineering
drwx------@   96 nandithagn  staff     3072  6 Dec  2025 Library
drwx------     7 nandithagn  staff      224  9 Apr  2026 Movies
drwx------+    6 nandithagn  staff      192 13 Sep  2025 Music
-rw-r--r--@    1 nandithagn  staff    14616  2 Sep 11:33 NANDITHA PROMT.docx
-rw-r--r--@    1 nandithagn  staff  2089356 29 Apr 22:46 PCL_FInal_Paper.pdf
drwx------+    6 nandithagn  staff      192 12 Aug 10:13 Pictures
drwxr-xr-x@    3 nandithagn  staff       96  3 Oct 22:23 Postman
drwxr-xr-x+    4 nandithagn  staff      128 22 Jul  2025 Public
-rw-r--r--@    1 nandithagn  staff   919432 22 Sep 19:34 RAKSHITH_R(26BTRCF037).pdf
-rw-r--r--@    1 nandithagn  staff   921871 22 Sep 19:28 RAKSHITH(Applied Pysics).pdf
-rw-r--r--@    1 nandithagn  staff   186078 21 Sep 14:38 WhatsApp Image 2026-09-21 at 14.25.48.jpeg
nandithagn@Nandithas-MacBook-Air ~ % date
Sun Oct 11 00:10:33 IST 2026
nandithagn@Nandithas-MacBook-Air ~ % whoami
nandithagn
nandithagn@Nandithas-MacBook-Air ~ % mkdir linux_practice
nandithagn@Nandithas-MacBook-Air ~ % cd linux_practice
nandithagn@Nandithas-MacBook-Air linux_practice % touch sample.txt
nandithagn@Nandithas-MacBook-Air linux_practice % ls -l
total 0
-rw-r--r--  1 nandithagn  staff  0 11 Oct 00:12 sample.txt
nandithagn@Nandithas-MacBook-Air linux_practice % pwd
/Users/nandithagn/linux_practice
nandithagn@Nandithas-MacBook-Air linux_practice % echo "Hello Linux" > sample.txt
nandithagn@Nandithas-MacBook-Air linux_practice % cat sample.txt
Hello Linux
nandithagn@Nandithas-MacBook-Air linux_practice % cp sample.txt copy.txt
nandithagn@Nandithas-MacBook-Air linux_practice % mv copy.txt renamed.txt
nandithagn@Nandithas-MacBook-Air linux_practice % ls -l
total 16
-rw-r--r--  1 nandithagn  staff  12 11 Oct 00:15 renamed.txt
-rw-r--r--  1 nandithagn  staff  12 11 Oct 00:15 sample.txt
nandithagn@Nandithas-MacBook-Air linux_practice %
```

Figure 26: 34 linux commands 11 to 15

Source file: 34_linux_commands_11_to_15.png

SCREENSHOT EVIDENCE 27 OF 58

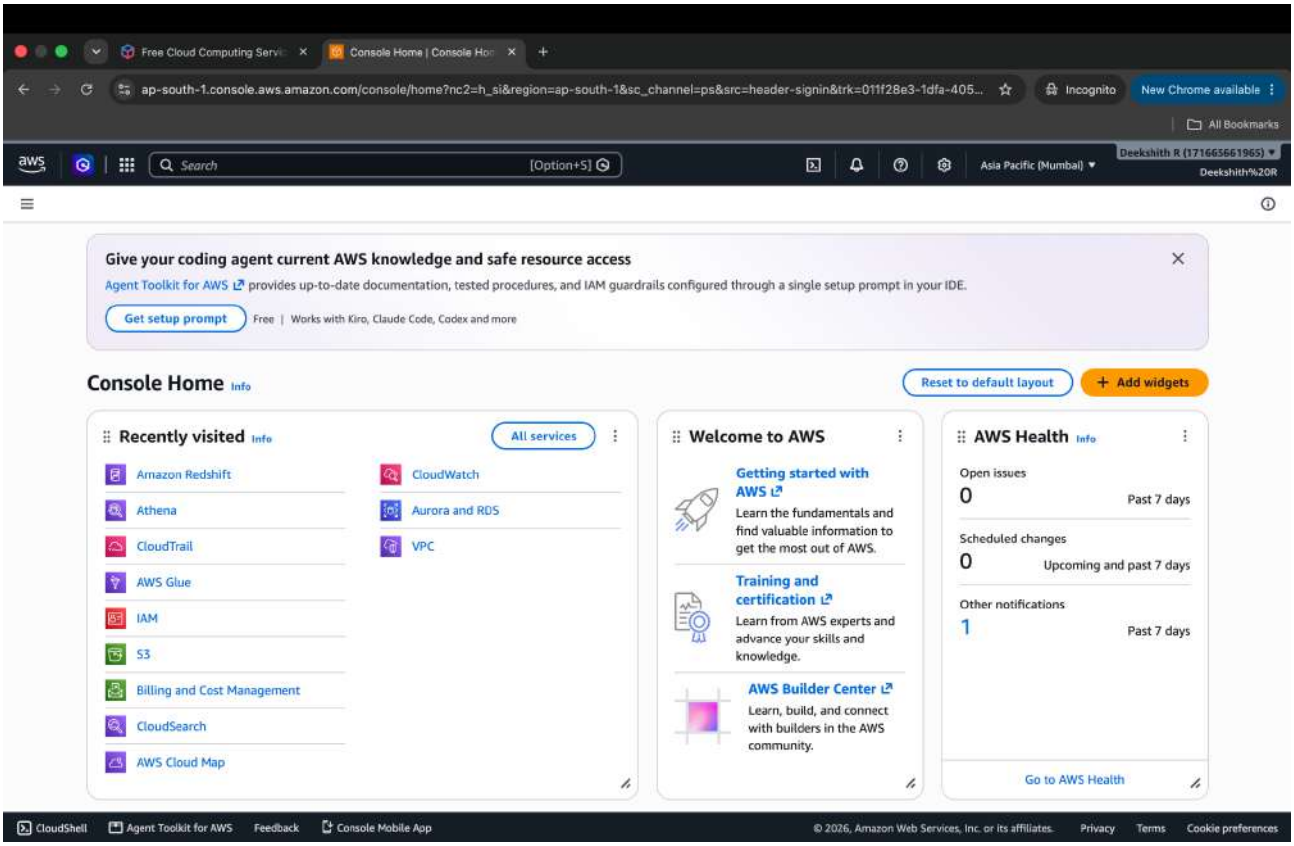


Figure 27: 37 AWS Account Region

Source file: 37_AWS_Account_Region.png

SCREENSHOT EVIDENCE 28 OF 58

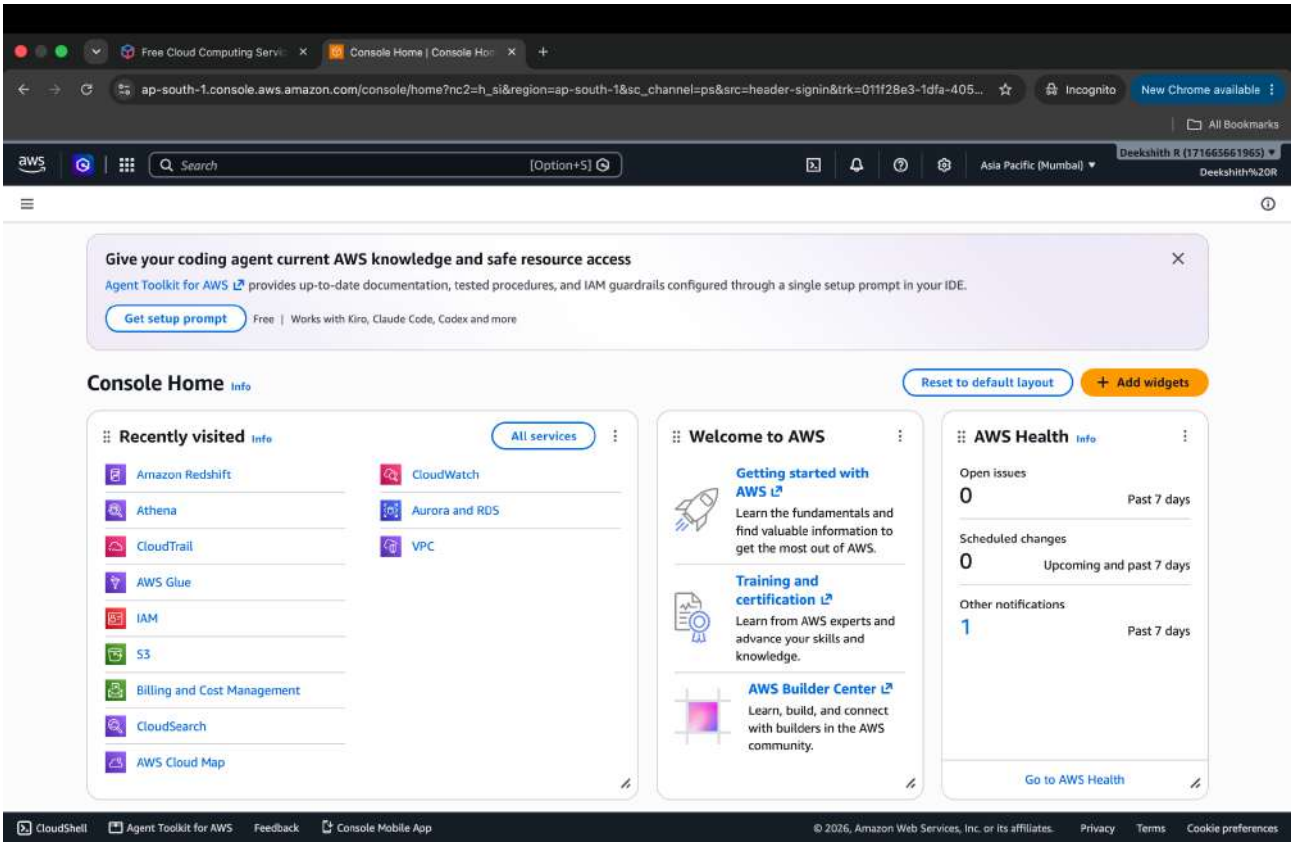


Figure 28: 37 AWS Account Region(1)

Source file: 37_AWS_Account_Region(1).png

SCREENSHOT EVIDENCE 29 OF 58

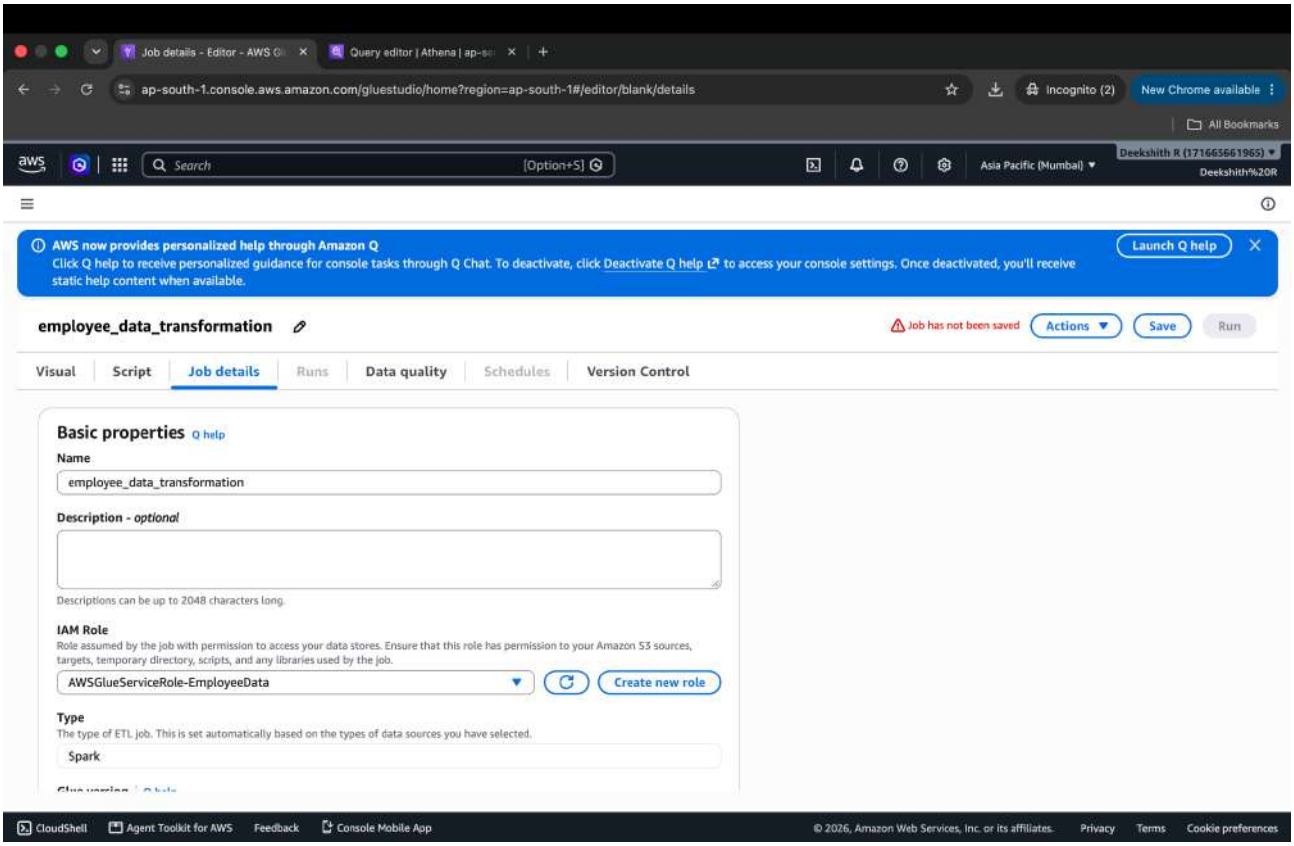


Figure 29: 54 Glue ETL Job Details

Source file: 54_Glue_ETL_Job_Details.png

SCREENSHOT EVIDENCE 30 OF 58



Figure 30: AWS console screenshot

Source file: image(20261010-172600).png

SCREENSHOT EVIDENCE 31 OF 58



Figure 31: AWS console screenshot

Source file: image(20261010-173641).png

SCREENSHOT EVIDENCE 32 OF 58



Figure 32: AWS console screenshot

Source file: image(20261010-173814).png

SCREENSHOT EVIDENCE 33 OF 58

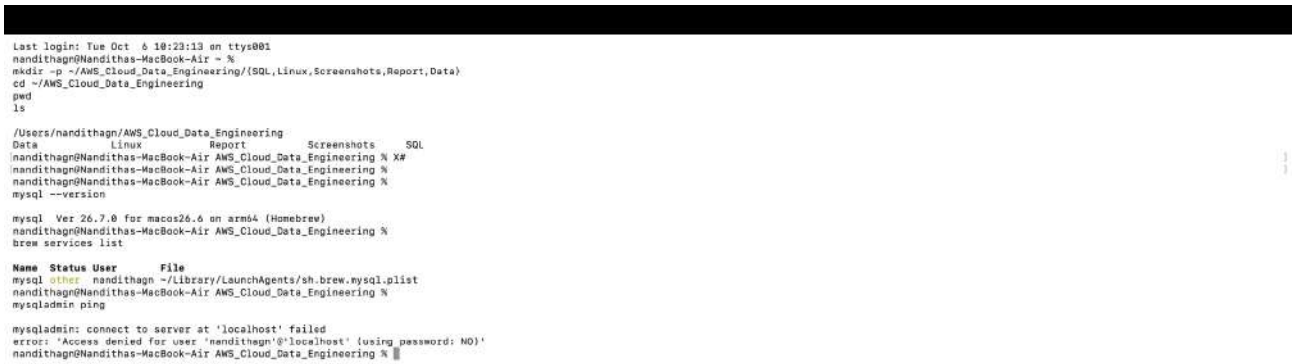


Figure 33: AWS console screenshot

Source file: image(20261010-174428).png

SCREENSHOT EVIDENCE 34 OF 58

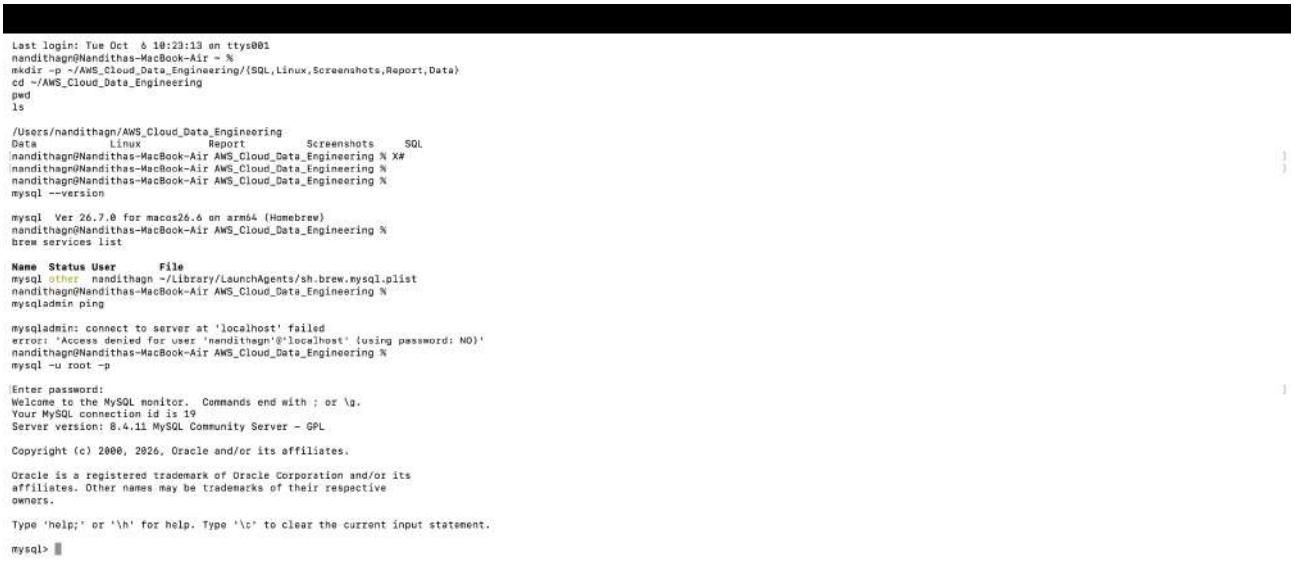


Figure 34: AWS console screenshot

Source file: image(20261010-174614).png

SCREENSHOT EVIDENCE 35 OF 58

```
+------------+----------------+
| city       | employee_count |
+------------+----------------+
| Bangalore |              4 |
| Hassan    |              3 |
| Hubli     |              3 |
| Mangalore |              2 |
| Mysore    |              3 |
+------------+----------------+
5 rows in set (0.001 sec)

mysql>
mysql> SELECT * FROM employees
    -> WHERE salary = (SELECT MAX(salary) FROM employees);
+-------------+------+------------+----------+-----------+
| employee_id | name | department | salary   | city      |
+-------------+------+------------+----------+-----------+
|         107 | Arjun | Finance    | 55000.00 | Bangalore |
+-------------+------+------------+----------+-----------+
1 row in set (0.001 sec)

mysql>
mysql> SELECT * FROM employees
    -> WHERE salary < 40000;
+-------------+-------+------------+----------+-----------+
| employee_id | name  | department | salary   | city      |
+-------------+-------+------------+----------+-----------+
|         102 | Diya  | HR         | 38000.00 | Mysore    |
|         105 | Vikram | Sales      | 35000.00 | Mangalore |
|         109 | Rahul | Sales      | 39000.00 | Hubli     |
|         113 | Manoj | Sales      | 36000.00 | Mysore    |
+-------------+-------+------------+----------+-----------+
4 rows in set (0.002 sec)

mysql>
mysql> SELECT MAX(salary) AS second_highest_salary
    -> FROM employees
    -> WHERE salary < (SELECT MAX(salary) FROM employees);
+------------------------+
| second_highest_salary |
+------------------------+
|               53000.00 |
+------------------------+
1 row in set (0.001 sec)

mysql>
mysql> SELECT *
    -> FROM employees
    -> WHERE salary > (SELECT AVG(salary) FROM employees);
+-------------+--------+------------+----------+-----------+
| employee_id | name   | department | salary   | city      |
+-------------+--------+------------+----------+-----------+
|         103 | Rohan  | Finance    | 52000.00 | Hubli     |
|         104 | Ananya | IT         | 48000.00 | Bangalore |
|         107 | Arjun  | Finance    | 55000.00 | Bangalore |
|         108 | Kavya  | IT         | 50000.00 | Mysore    |
|         110 | Sneha  | Finance    | 47000.00 | Hassan    |
|         111 | Kiran  | IT         | 46000.00 | Mangalore |
|         114 | Pooja  | Finance    | 53000.00 | Hubli     |
|         115 | Aditya | IT         | 49000.00 | Hassan    |
+-------------+--------+------------+----------+-----------+
8 rows in set (0.001 sec)

mysql>
```

Figure 35: AWS console screenshot

Source file: image(20261010-183141).png

SCREENSHOT EVIDENCE 36 OF 58



Figure 36: AWS console screenshot

Source file: image(20261010-184654).png

SCREENSHOT EVIDENCE 37 OF 58

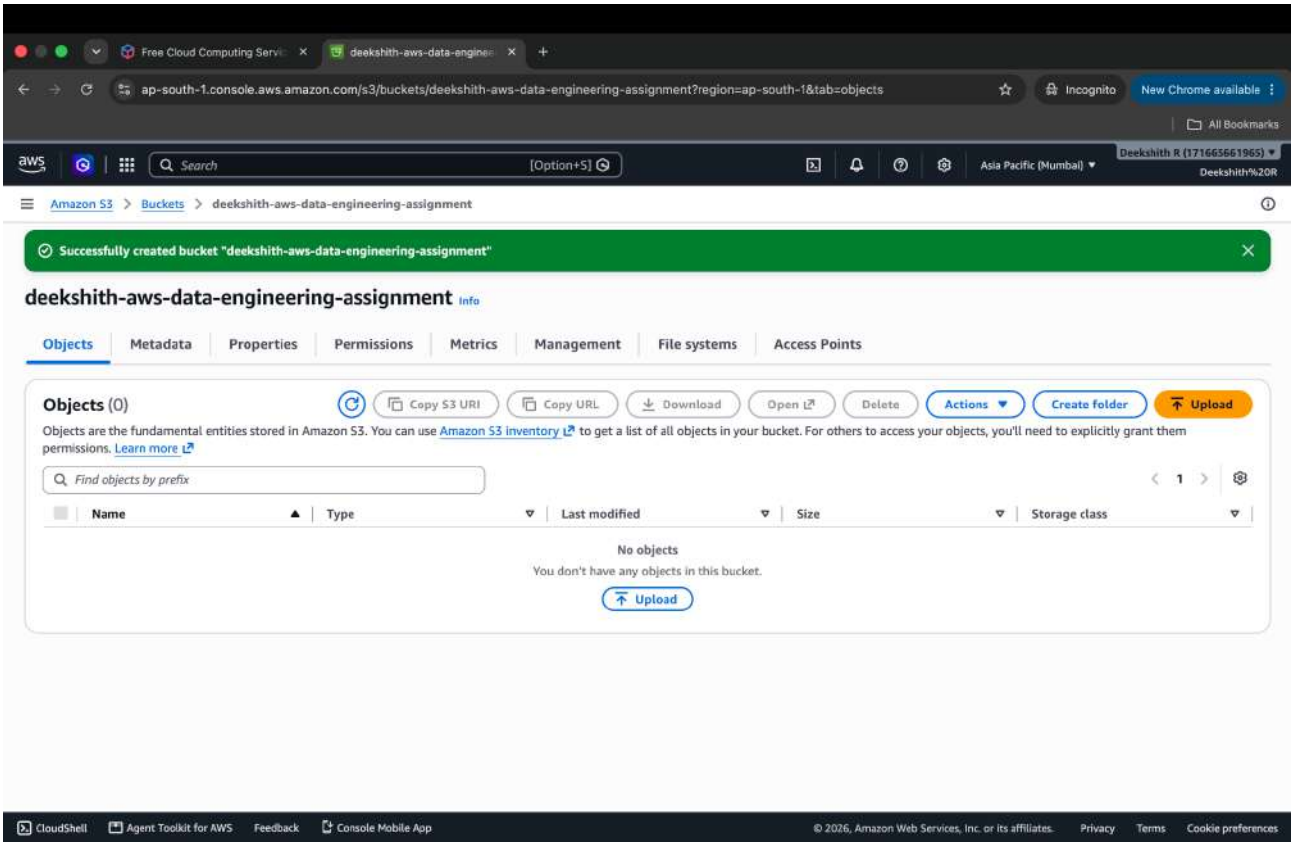


Figure 37: AWS console screenshot

Source file: image(20261010-185612).png

SCREENSHOT EVIDENCE 38 OF 58

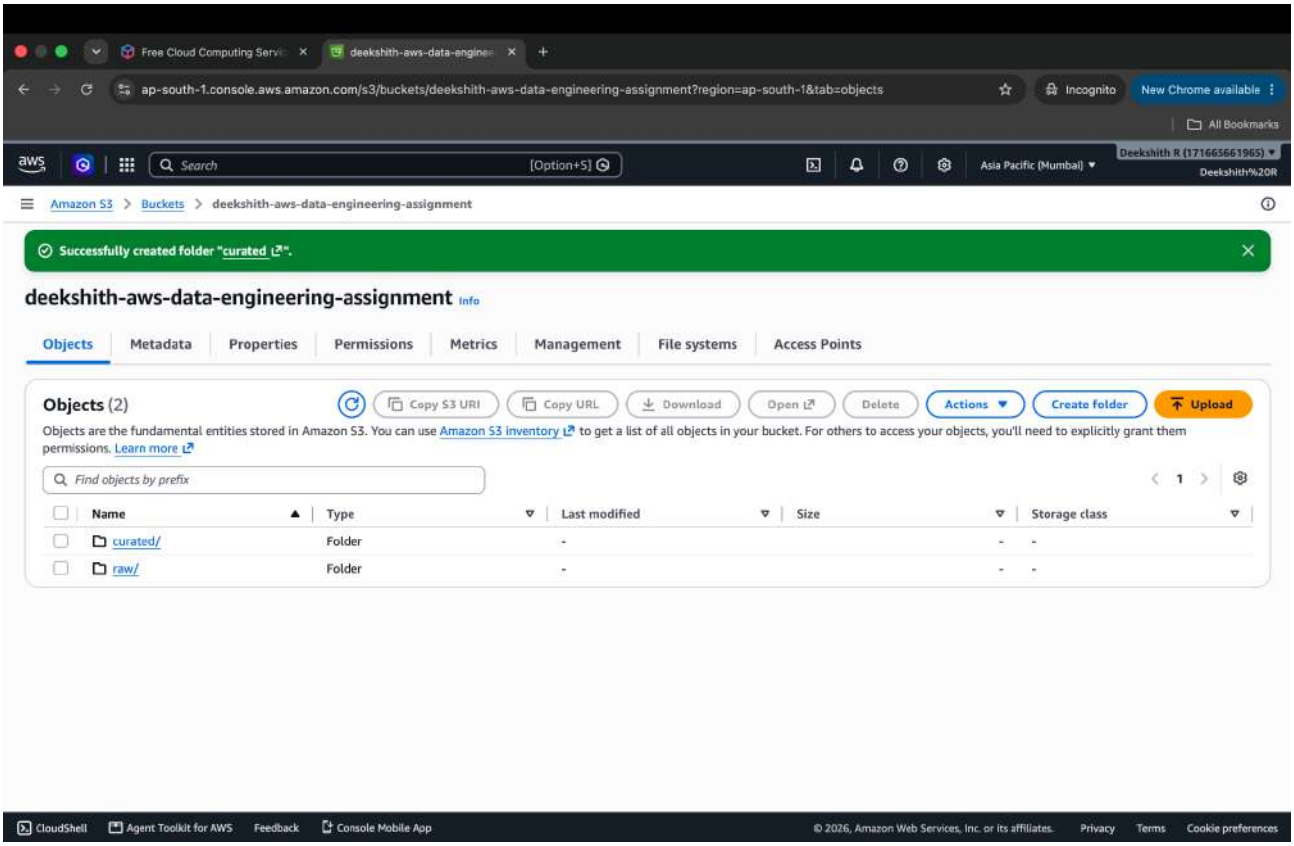


Figure 38: AWS console screenshot

Source file: image(20261010-185719).png

SCREENSHOT EVIDENCE 39 OF 58

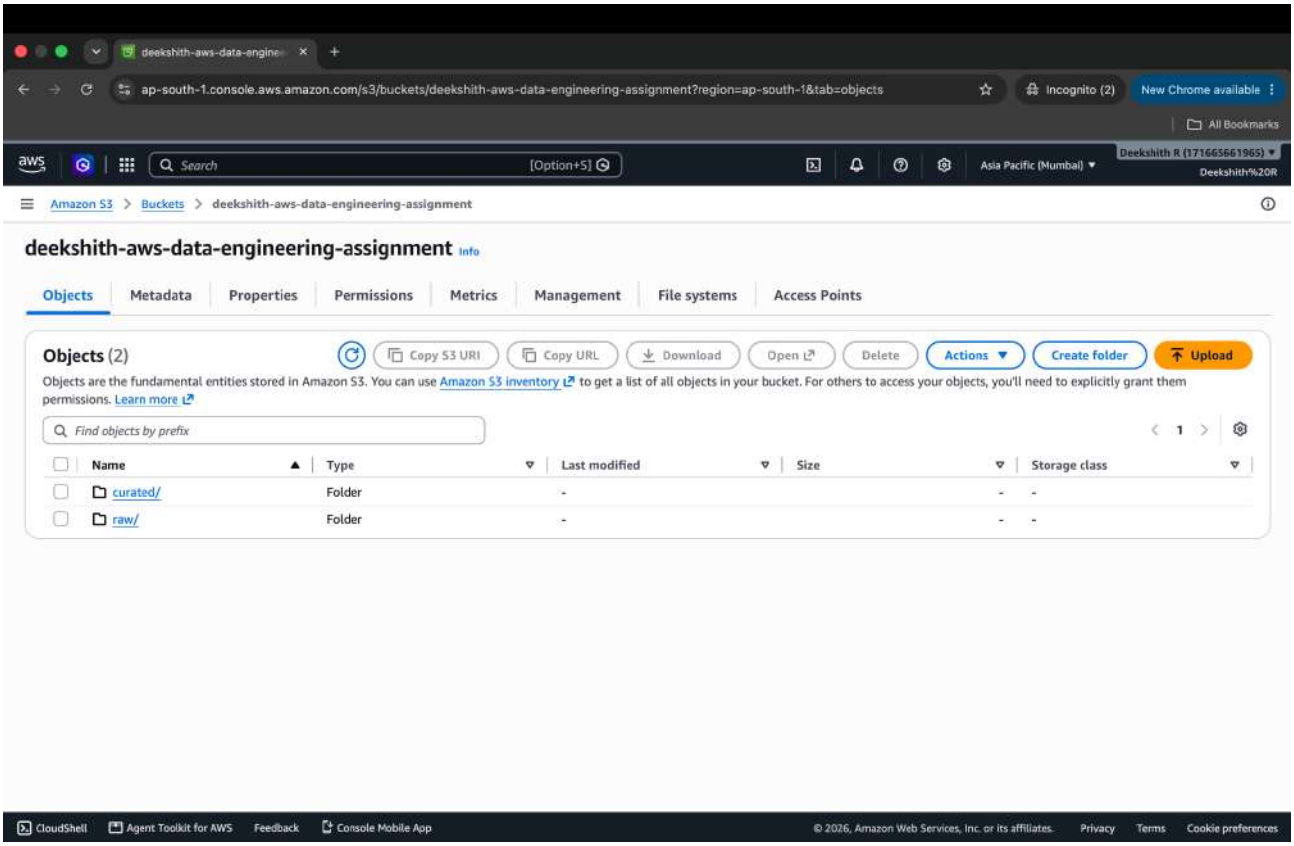


Figure 39: AWS console screenshot

Source file: image(20261012-054713).png

SCREENSHOT EVIDENCE 40 OF 58

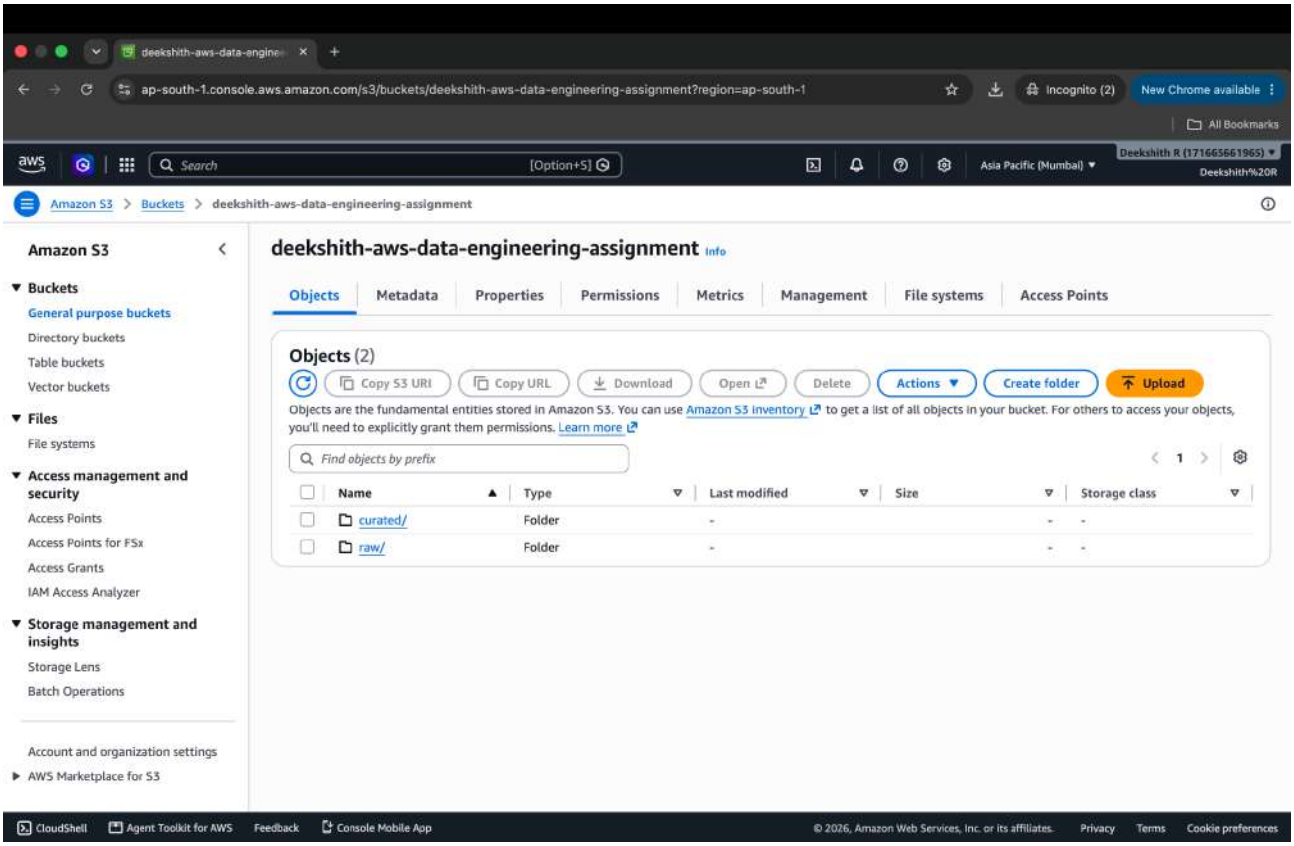


Figure 40: AWS console screenshot

Source file: image(20261012-054955).png

SCREENSHOT EVIDENCE 41 OF 58

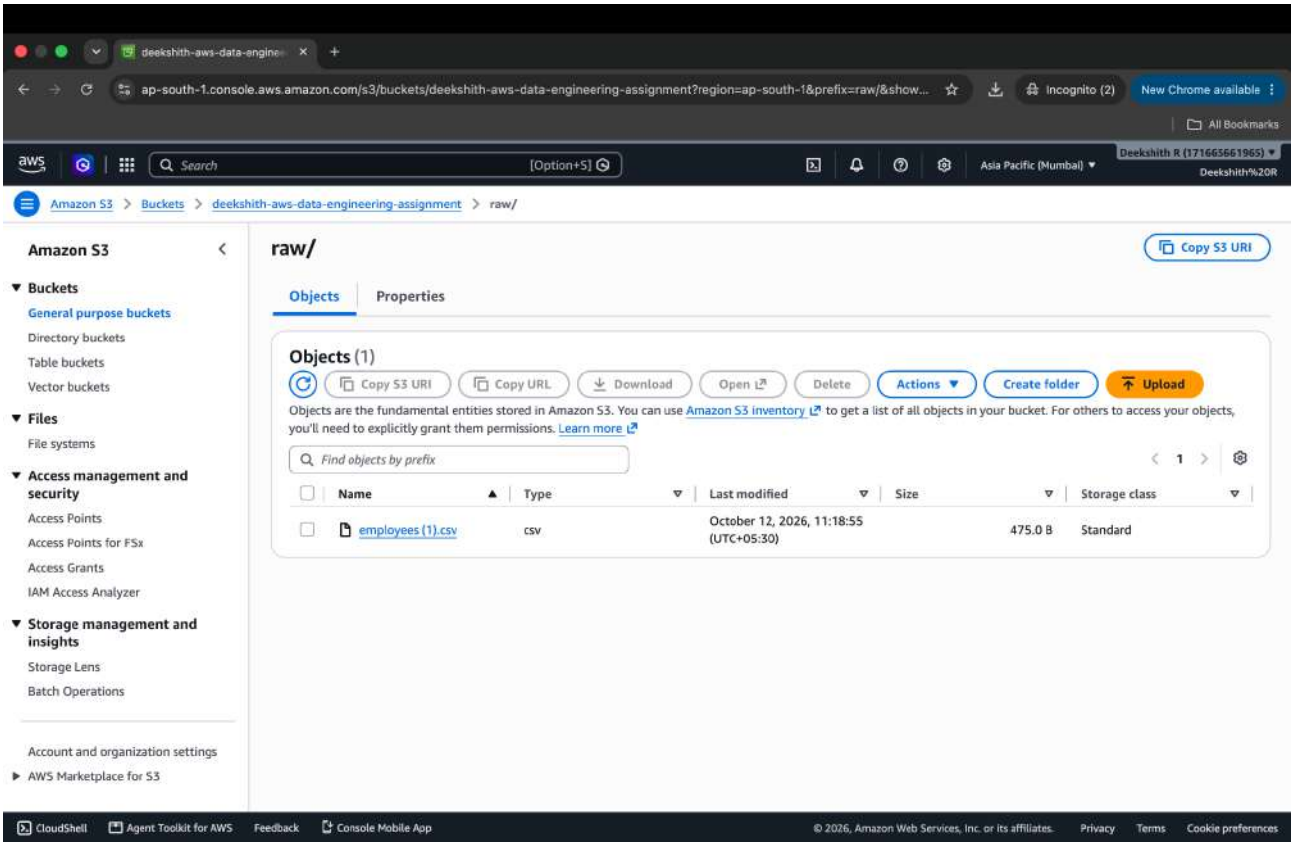


Figure 41: AWS console screenshot

Source file: image(20261012-055022).png

SCREENSHOT EVIDENCE 42 OF 58

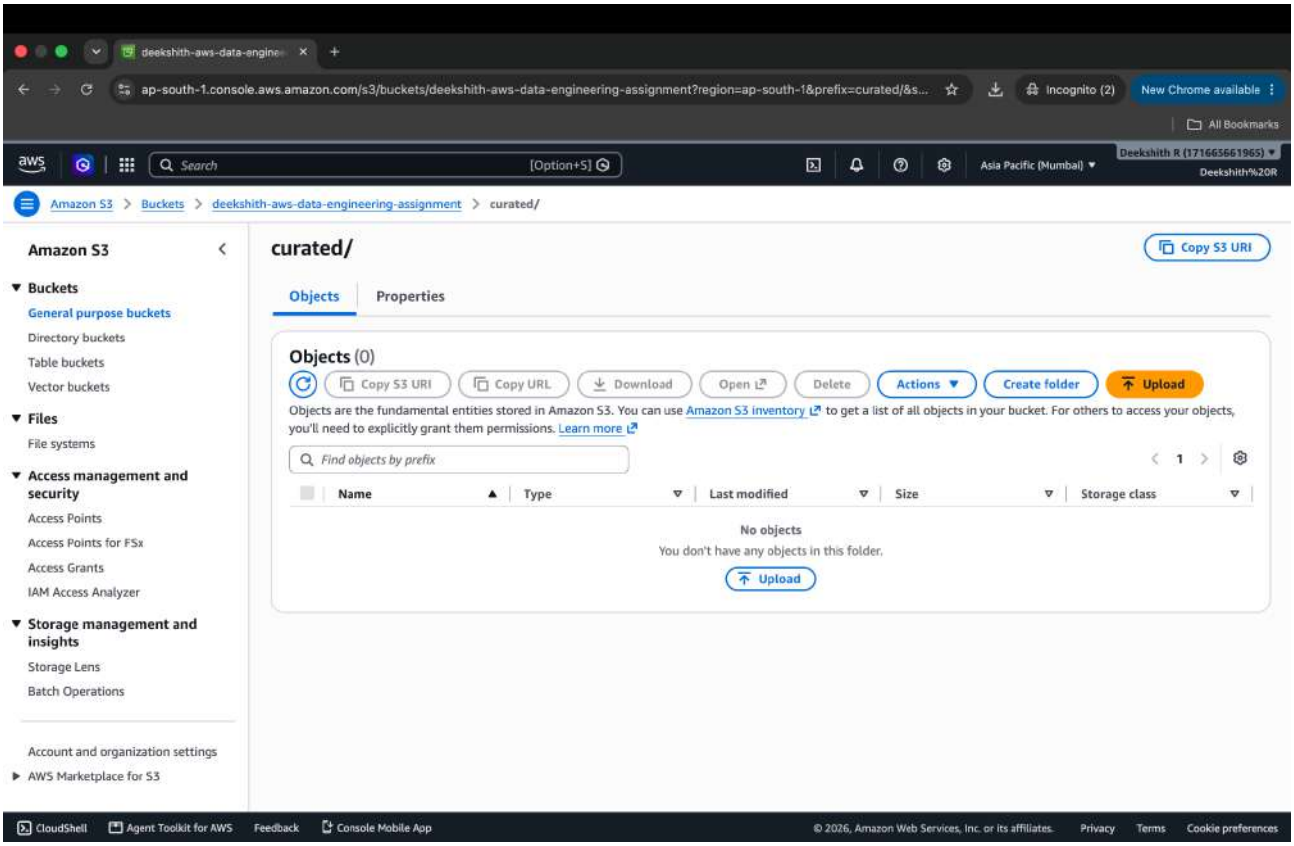


Figure 42: AWS console screenshot

Source file: image(20261012-055217).png

SCREENSHOT EVIDENCE 43 OF 58

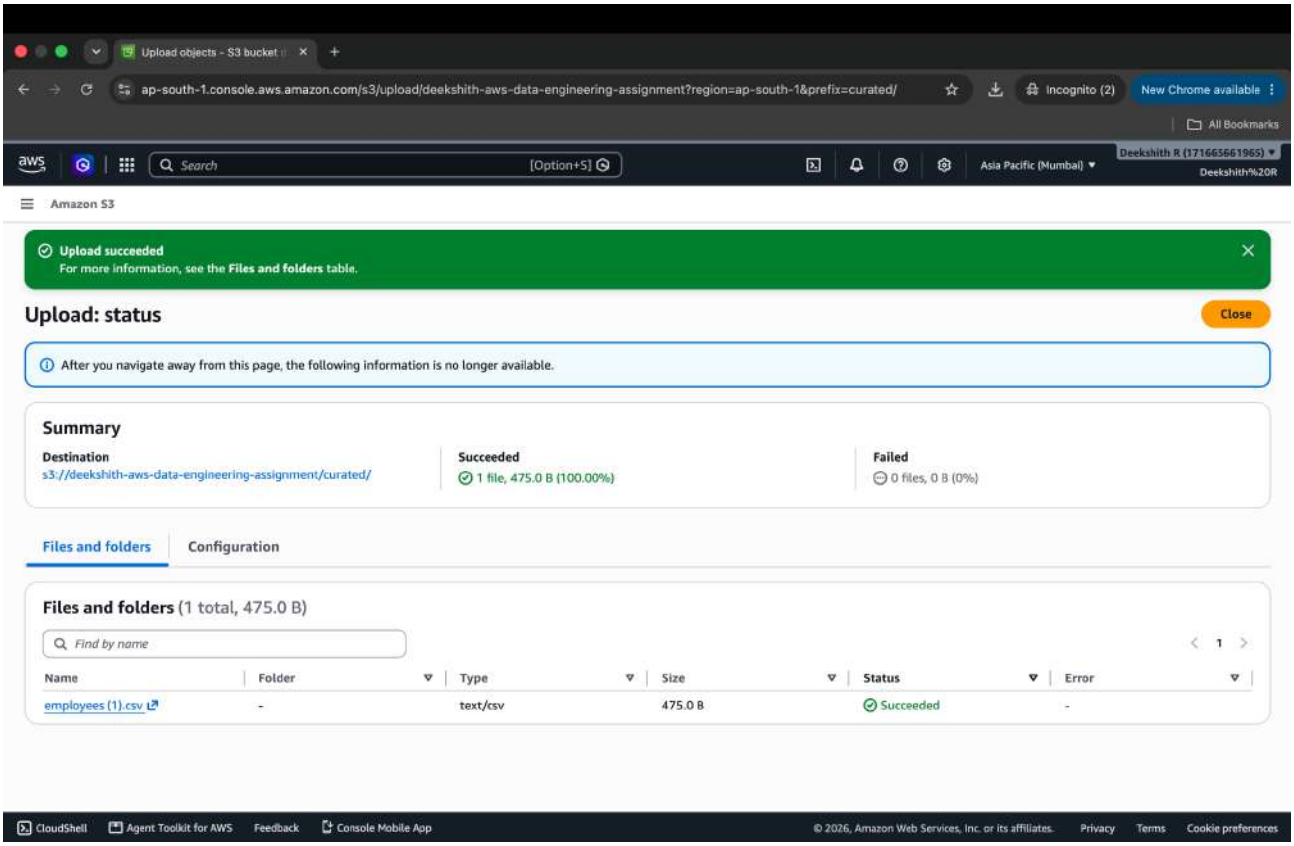


Figure 43: AWS console screenshot

Source file: image(20261012-055356).png

SCREENSHOT EVIDENCE 44 OF 58

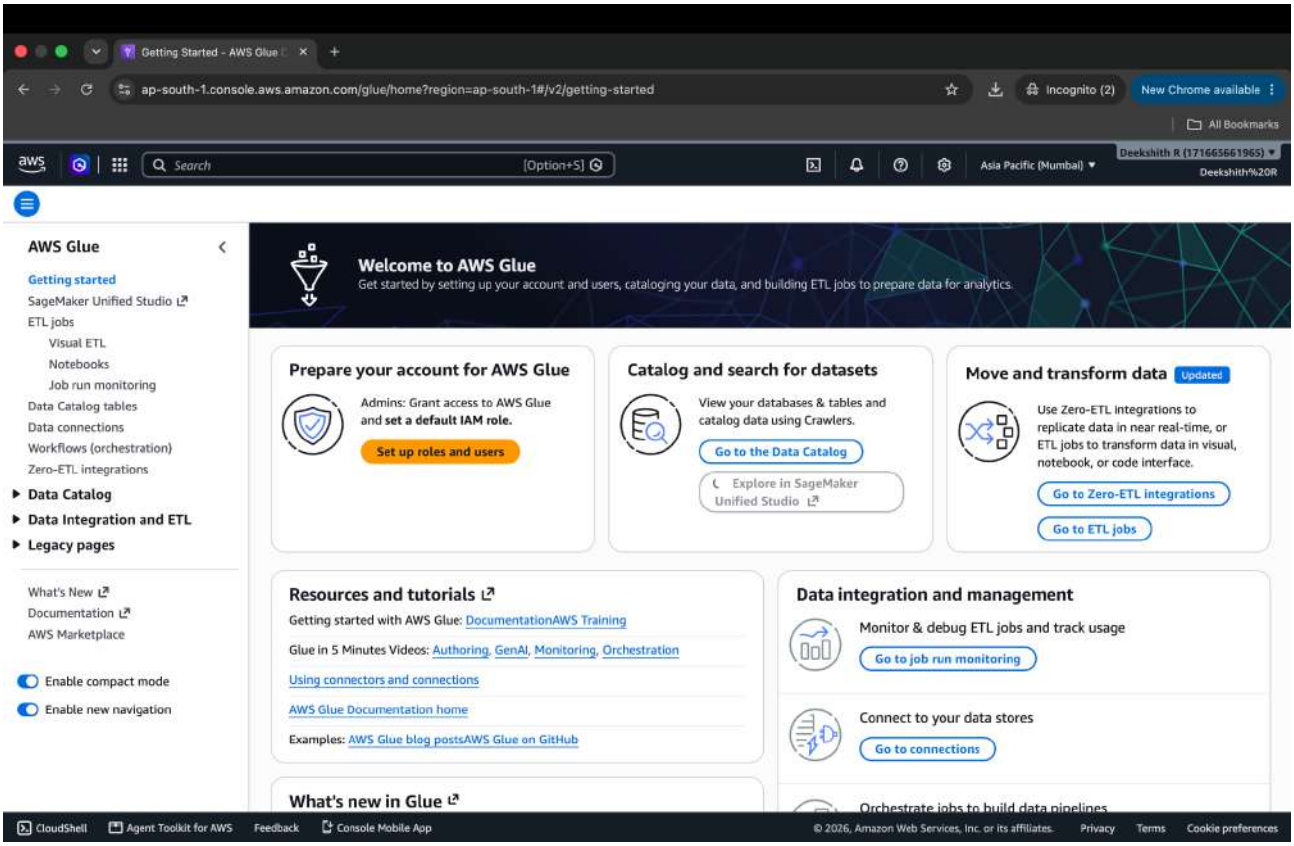


Figure 44: AWS console screenshot

Source file: image(20261012-055705).png

SCREENSHOT EVIDENCE 45 OF 58

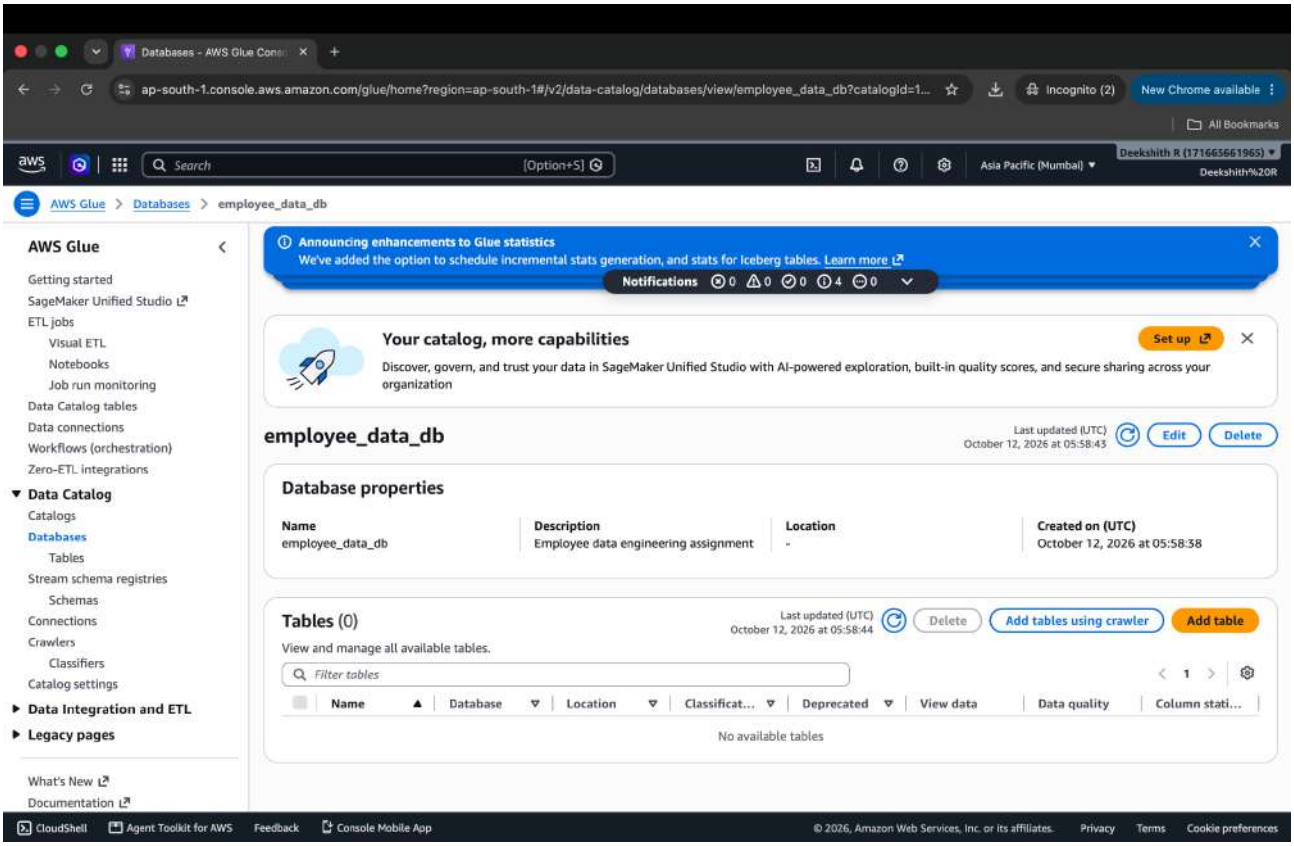


Figure 45: AWS console screenshot

Source file: image(20261012-055901).png

SCREENSHOT EVIDENCE 46 OF 58

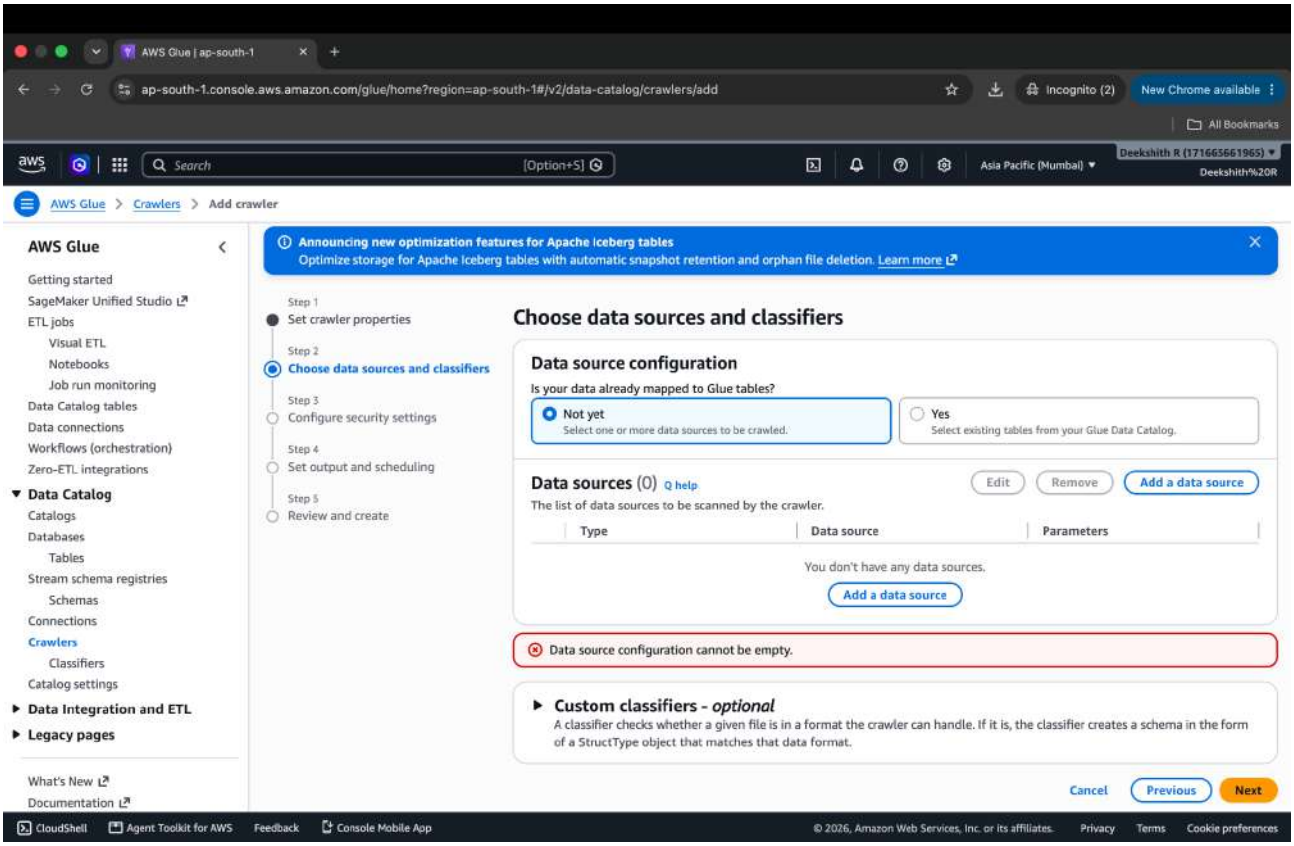


Figure 46: AWS console screenshot

Source file: image(20261012-060050).png

SCREENSHOT EVIDENCE 47 OF 58

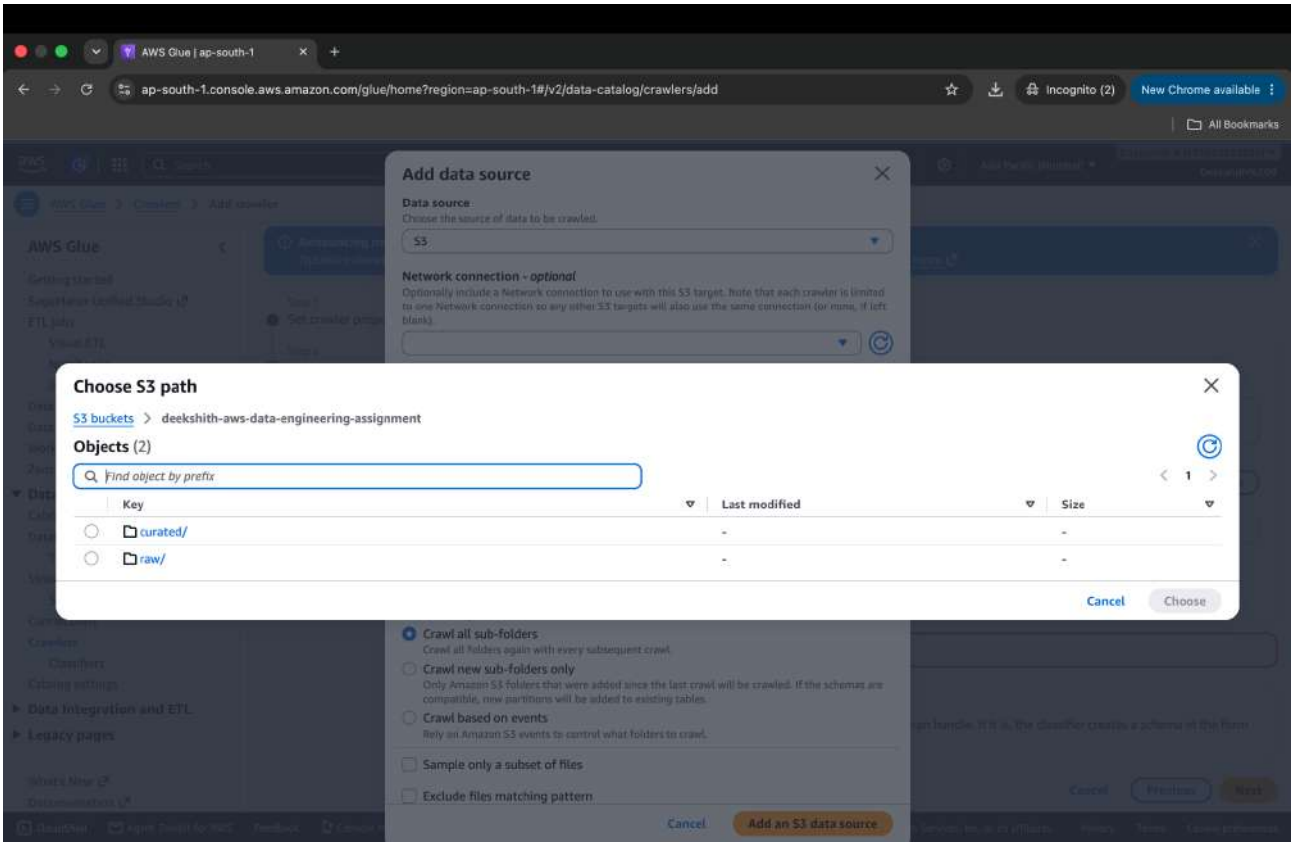


Figure 47: AWS console screenshot

Source file: image(20261012-060141).png

SCREENSHOT EVIDENCE 48 OF 58

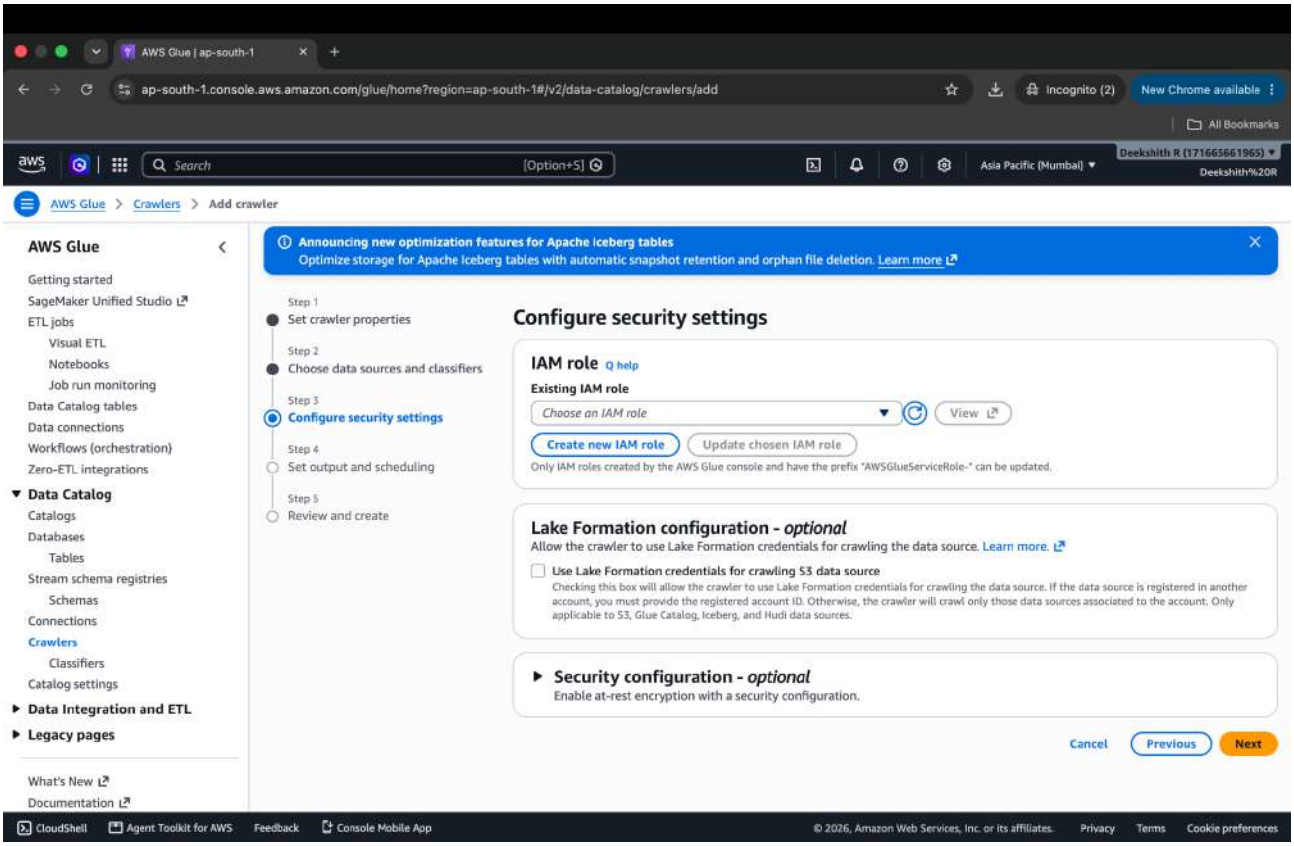


Figure 48: AWS console screenshot

Source file: image(20261012-060237).png

SCREENSHOT EVIDENCE 49 OF 58

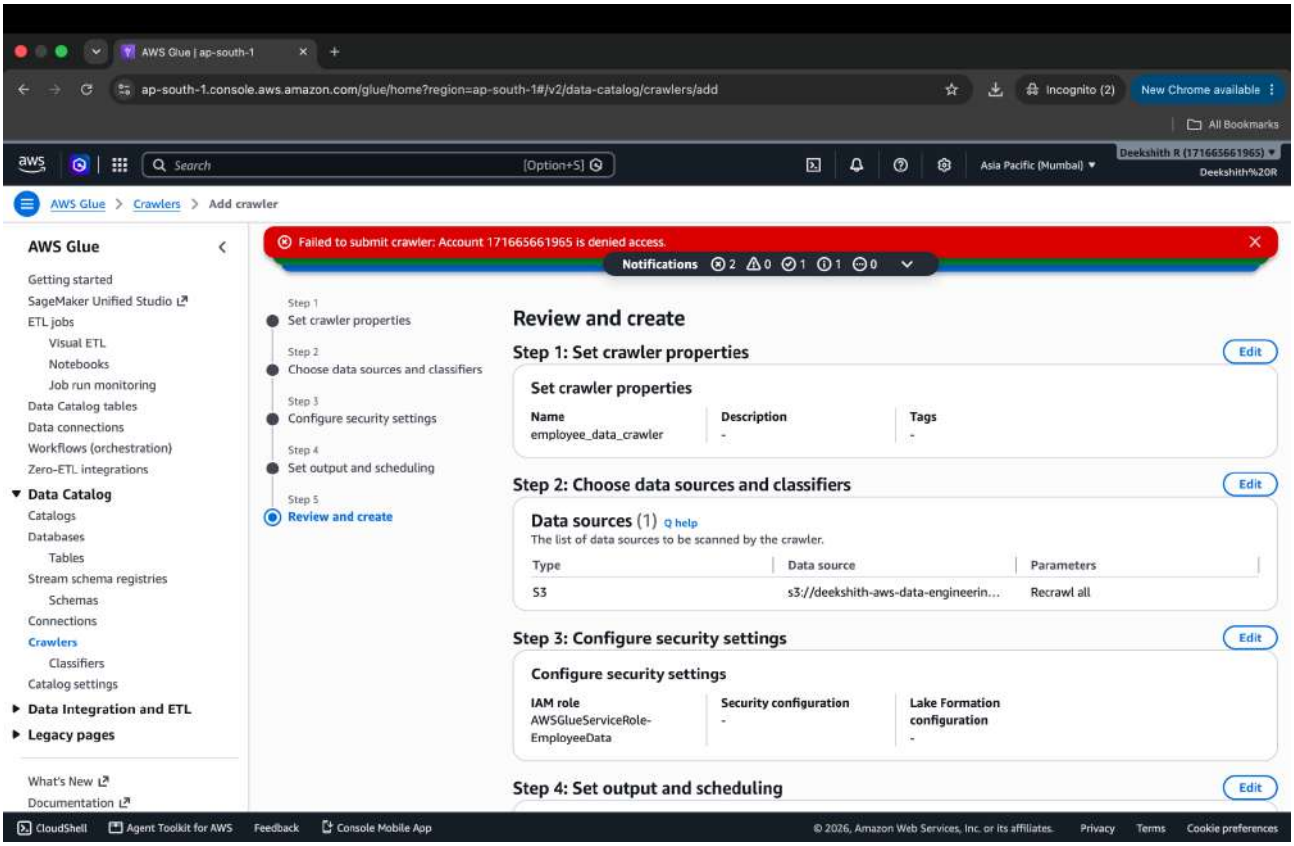


Figure 49: AWS console screenshot

Source file: image(20261012-060403).png

SCREENSHOT EVIDENCE 50 OF 58

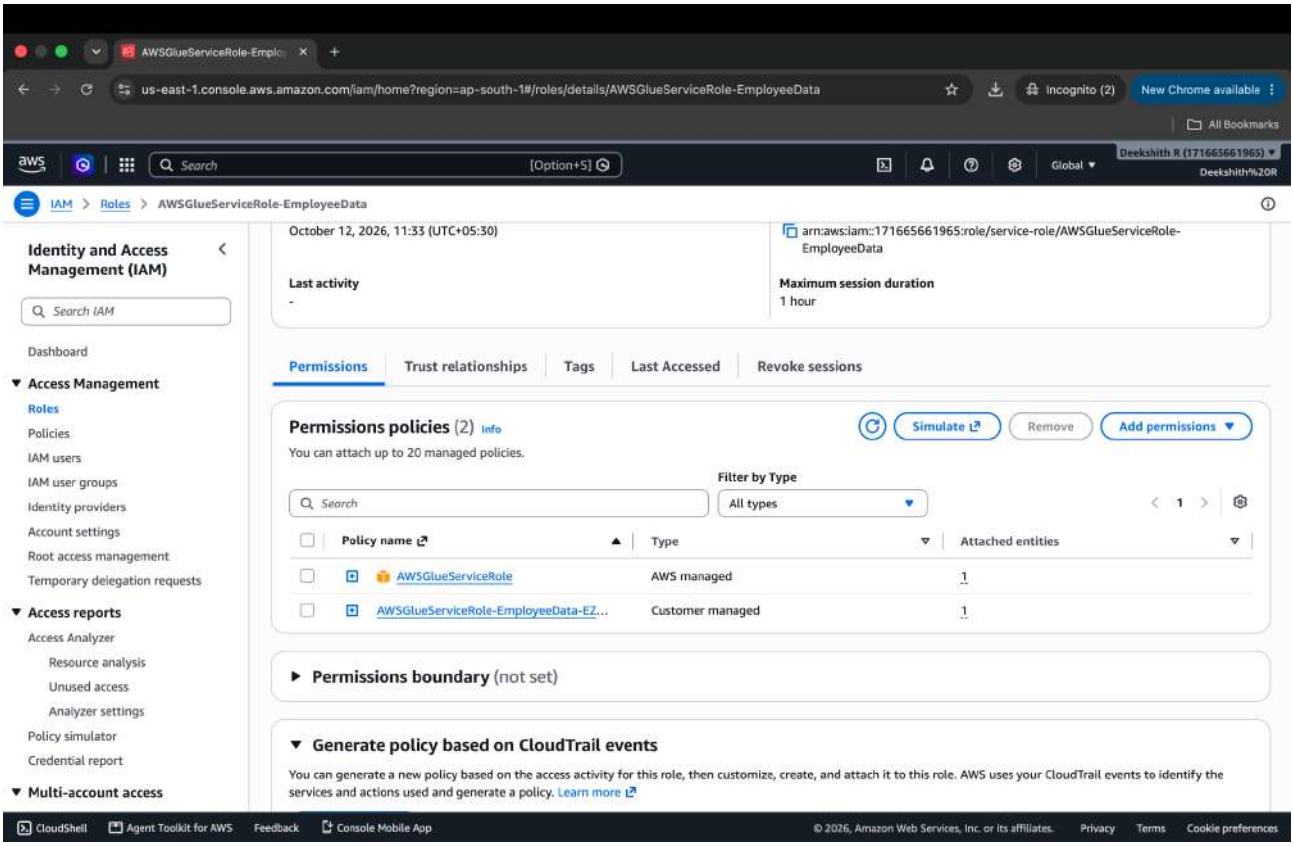


Figure 50: AWS console screenshot

Source file: image(20261012-060646).png

SCREENSHOT EVIDENCE 51 OF 58

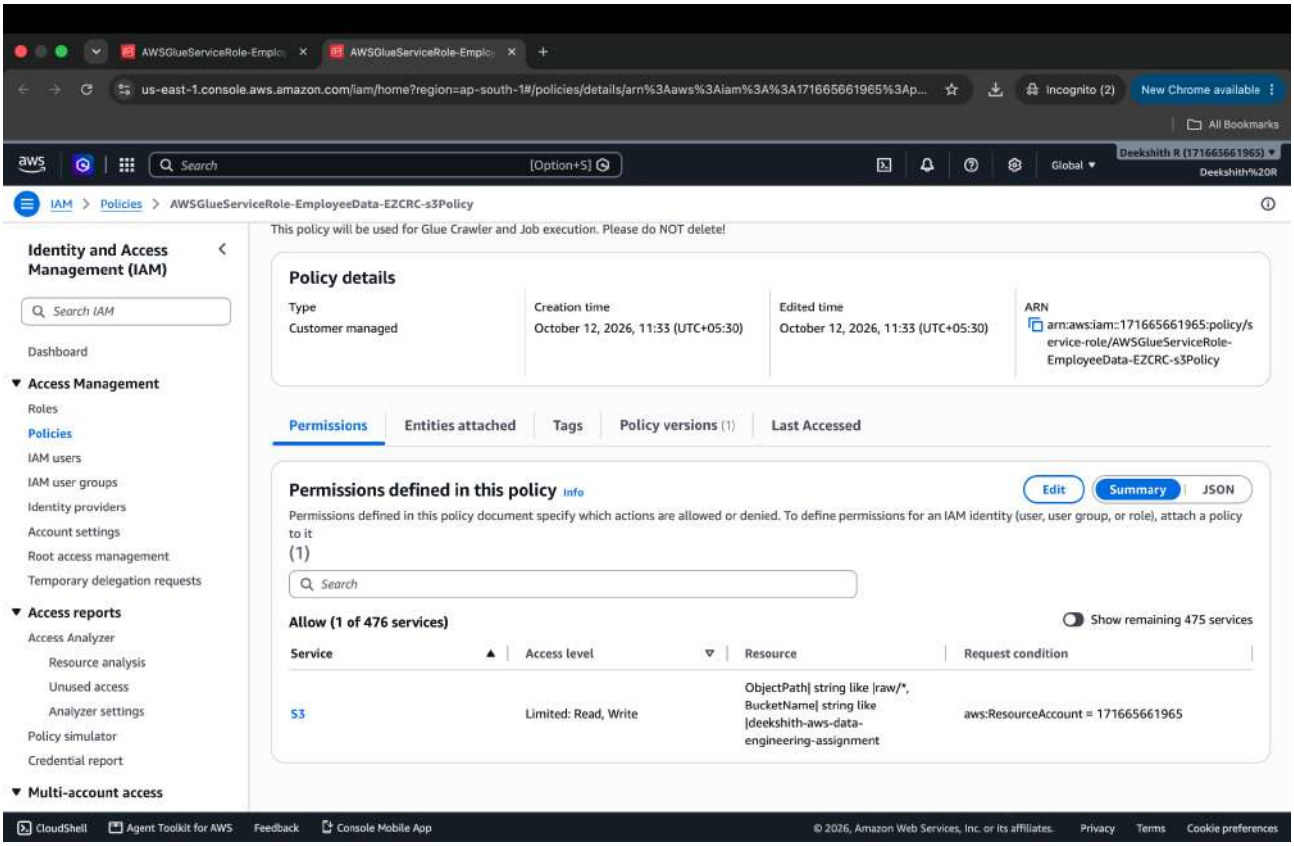


Figure 51: AWS console screenshot

Source file: image(20261012-060750).png

SCREENSHOT EVIDENCE 52 OF 58

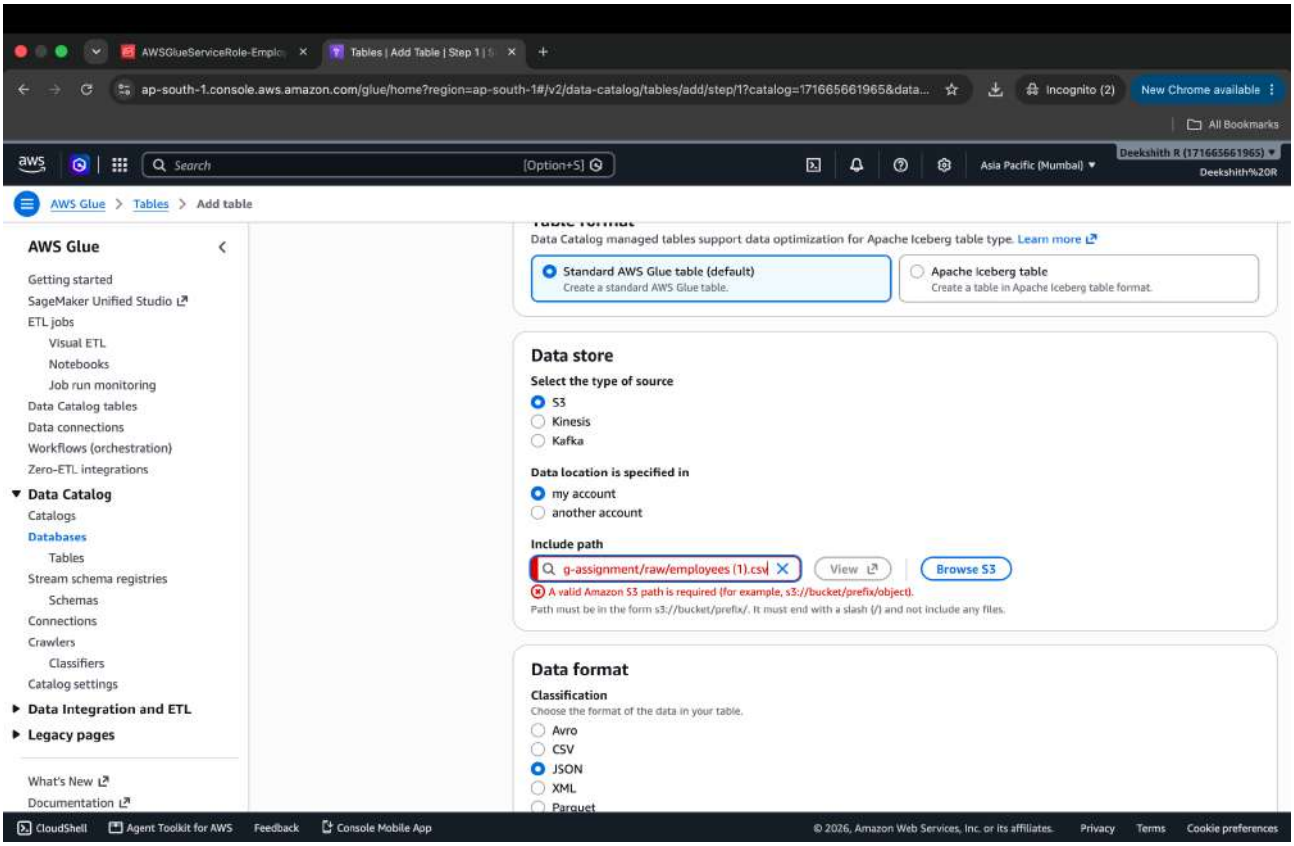


Figure 52: AWS console screenshot

Source file: image(20261012-061115).png

SCREENSHOT EVIDENCE 53 OF 58

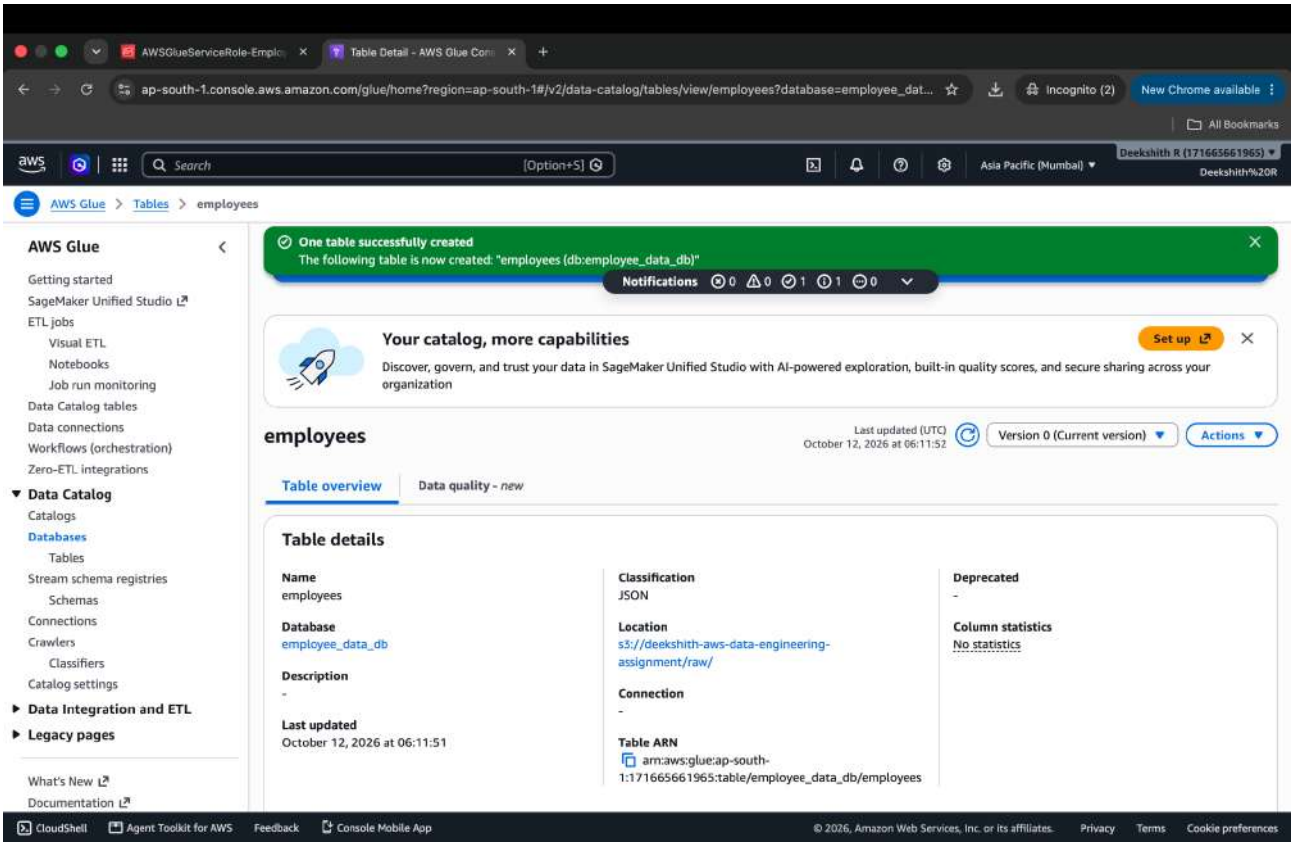


Figure 53: AWS console screenshot

Source file: image(20261012-061210).png

SCREENSHOT EVIDENCE 54 OF 58

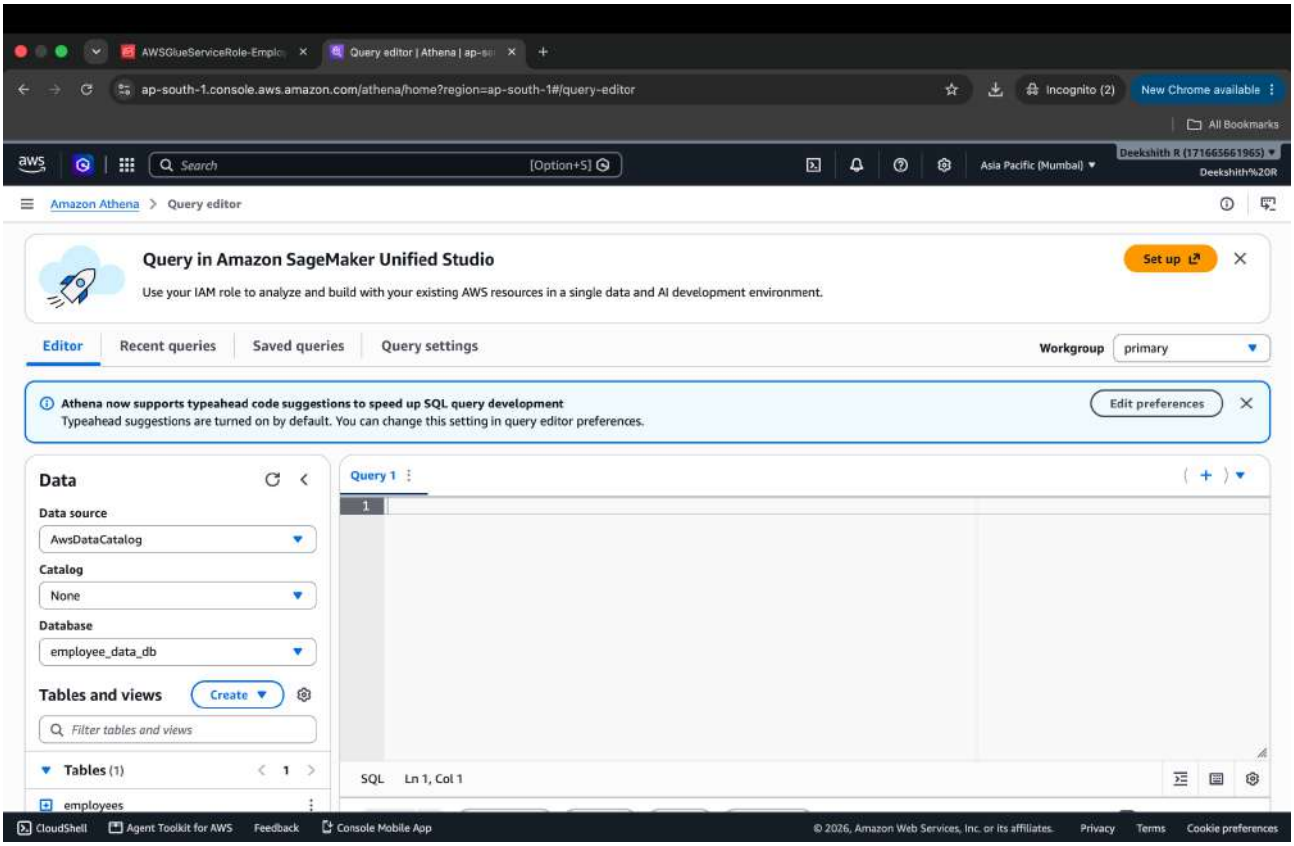


Figure 54: AWS console screenshot

Source file: image(20261012-061338).png

SCREENSHOT EVIDENCE 55 OF 58

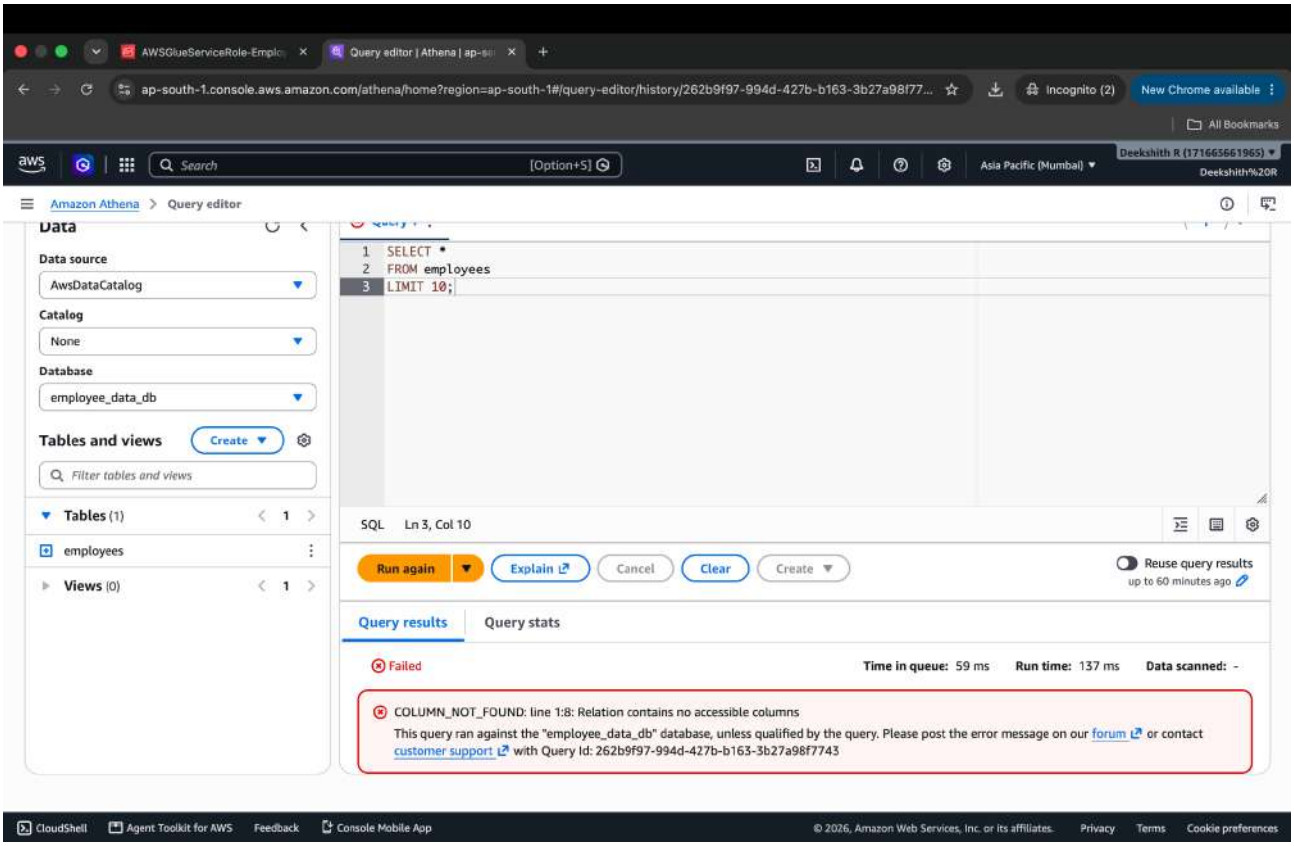


Figure 55: AWS console screenshot

Source file: image(20261012-061417).png

SCREENSHOT EVIDENCE 56 OF 58

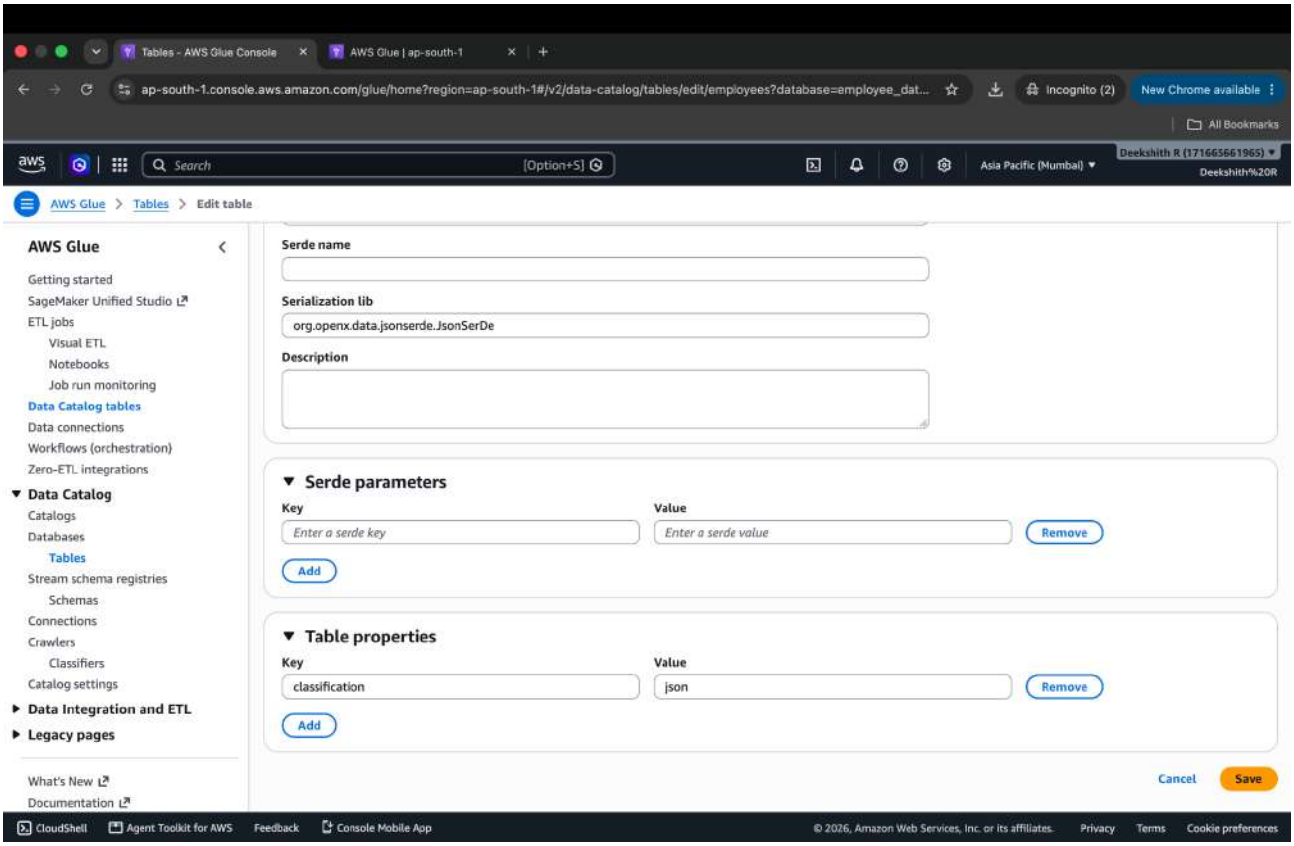


Figure 56: AWS console screenshot

Source file: image(20261012-061648).png

SCREENSHOT EVIDENCE 57 OF 58

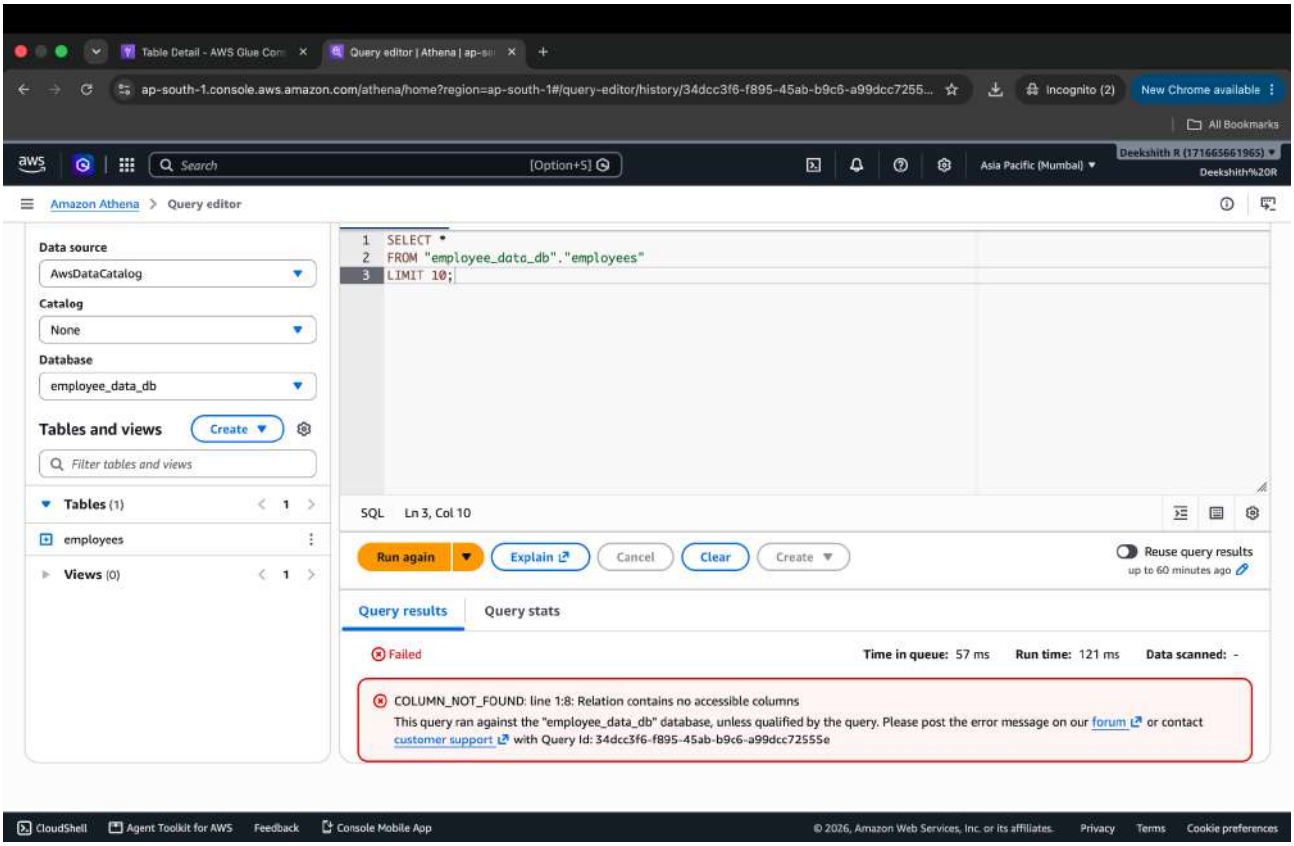


Figure 57: AWS console screenshot

Source file: image(20261012-061934).png

SCREENSHOT EVIDENCE 58 OF 58

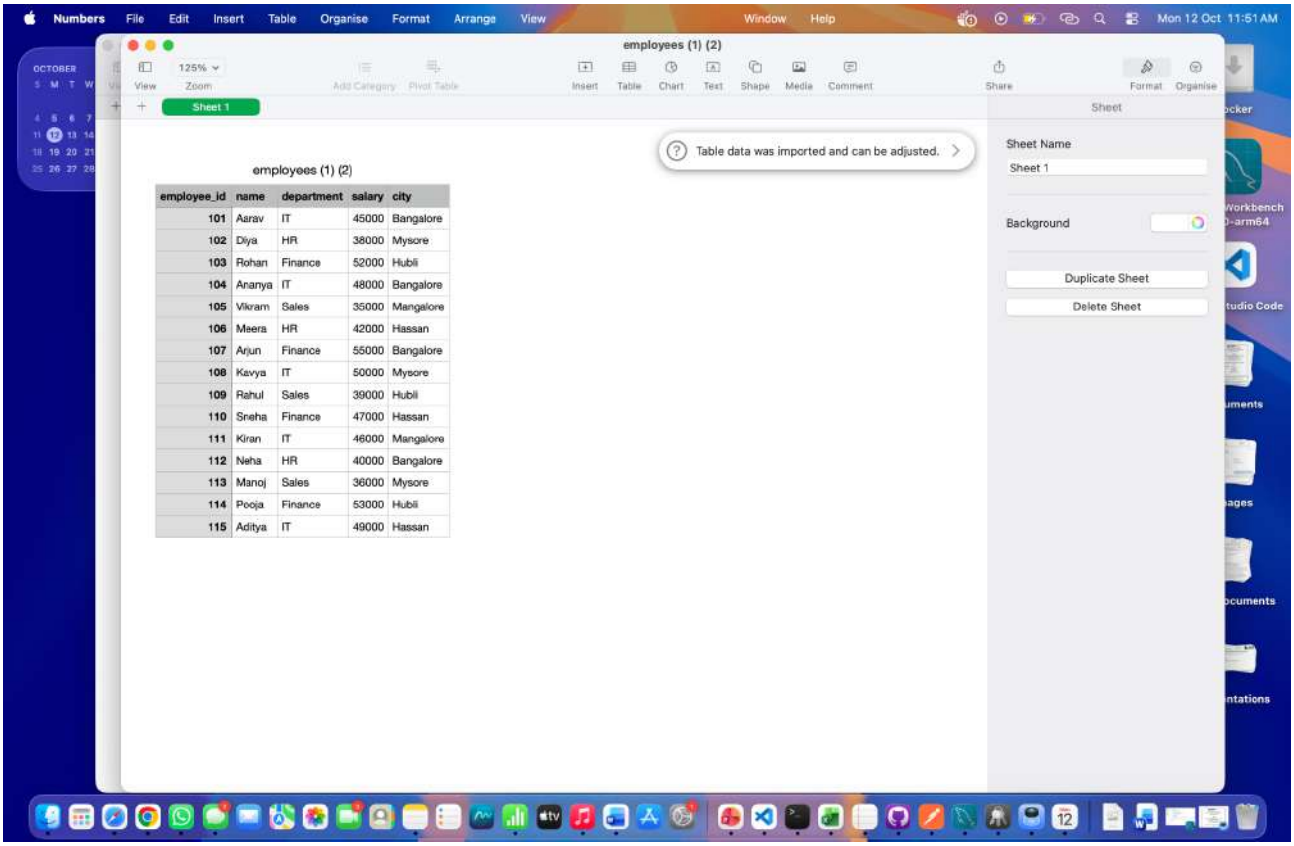


Figure 58: AWS console screenshot

Source file: image(20261012-062149).png